Mandatory Deadlines and Human Capital Accumulation: Evidence from a Natural Experiment in Higher Education

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In flexible academic environments, students often face weak incentives for sustained effort, which can facilitate procrastination and delay the accumulation of human capital. This study exploits a natural experiment at a German university where two programs introduced course-completion deadlines in 2016—one early and targeted (first-semester), the other delayed and comprehensive (second-year). Using administrative panel data covering over 11,500 students and difference-in-differences (DiD) estimation, I find that early deadlines increased first-year credits by 18% and graduation rates by 7 percentage points. The delayed deadline generated similar effect magnitudes—12% increase in long-term earned credits and a 5.7 percentage-point rise in on-time graduation, with effects reaching statistical significance in propensity-score matched and synthetic DiD specifications. Dropout rates and GPA remained unchanged in both regimes, indicating improvements stem from increased effort rather than selection. Broadly distributed effects across ability levels suggest these deadlines serve as universal pacing tools, enabling institutions to generate academic momentum and improve human capital accumulation without sacrificing equity.

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1. Introduction

A central question in the design of higher education institutions is how to structure academic environments to maximize human capital accumulation. Theoretically, institutional flexibility allows students to allocate their effort over time optimally; in practice, however, behavioral frictions often disrupt this optimization (Lavecchia, Liu, & Oreopoulos, 2016). In Germany, these frictions can contribute to sizable inefficiencies: roughly 30% of bachelor's students drop out after an average of 4.7 semesters, and only about one-third of graduates finish on time (Heublein et al., 2017; Statistisches Bundesamt, 2024). These outcomes are often linked to the system's strong emphasis on student autonomy, reflected in tuition-free enrollment and relatively flexible examination regulations (Bäulke and Dresel, 2023). Lacking immediate constraints, students face little cost for delay, making them structurally vulnerable to drifting off track.

In response to sub-optimal student outcomes, universities have begun experimenting with mandatory course-completion deadlines to enforce consistent engagement. Such deadlines target behavioral frictions by serving as institutional commitment devices. By attaching immediate consequences to inaction, they force students to internalize future costs and sustain effort (Ariely & Wertenbroch, 2002; Kaur et al., 2015). However, the empirical evidence on such constraints is mixed: while some interventions successfully increase effort, others fail to improve completion rates or yield null results (Burger, Charness & Lynham, 2011). This variation suggests that the effectiveness of deadlines likely depends on specific design parameters, such as timing, scope, and enforcement strictness.

This paper studies the causal effects of course-completion deadlines using a natural experiment at a German university of applied sciences. In 2016, two large study programs introduced new deadline regimes, while other programs at the same institution continued under unchanged regulations. The reforms differed sharply in design. Business Administration implemented an early, targeted, and high-stakes deadline affecting two foundational first-semester modules. Social Work adopted a delayed, comprehensive, and lower-intensity deadline requiring completion of all first-year coursework by the end of the second year. Crucially, both policies function as mandatory attempt requirements; failure to meet the deadline results in a counted failed attempt rather than immediate program dismissal, preserving remaining retake opportunities. This within-institution variation across programs and cohorts allows me to examine how the timing and scope of deadlines shape student responses.

I utilize administrative panel data covering over 11,500 students across seven cohorts (2012–2018) and employ a difference-in-differences (DiD) strategy that uses variation in course completion deadlines across programs and cohorts. As additional robustness layers, I re-estimate the effects using propensity-score matching (PSM) to improve covariate balance and synthetic difference-in-differences (SDiD) (Arkhangelsky et al., 2021) to construct optimized synthetic counterfactuals, ensuring the findings are robust to demographic compositional shifts and potential trend heterogeneity.

The results show that mandatory deadlines can meaningfully shift academic trajectories, particularly when imposed early and with clear consequences. In the BA program, treated cohorts earned roughly 7 additional credits in their first year (around 1.4 additional courses), a significant 18% increase over pre-reform cohorts, indicating greater academic momentum by the time the deadline first bound. Short-run gains were achieved without any decline in GPA and without an increase in dropout rate. These results indicate that students responded to the reform by raising total study effort, not by diluting performance or exiting early.

This early acceleration translated into sustained long-run improvements. By semester seven, treated students accumulated an additional 21.9 credits—a 17% gain over pre-reform cohorts. This proportional increase mirrors the short-run effect, demonstrating that the reform initiated a self-reinforcing trajectory rather than merely shifting effort forward in time. These gains are robust across specifications: while marginally significant in the baseline model ($p=0.054$), they are strongly significant in both propensity-score matched and SDiD specifications ($p=0.001$). Similarly, cumulative dropout fell by 5.5 percentage points. While this estimate is less precise in the conservative baseline, the matched specification indicates a statistically significant reduction ($p=0.058$), suggesting that improved outcomes were achieved in larger, more heterogeneous cohorts and ruling out mechanical selection effects. Finally, GPA remained stable, confirming that students maintained academic quality while accelerating their pace.

Finally, the reforms substantially enhanced degree completion rates. Graduation within nine semesters rose by 7.0 percentage points, while on-time graduation increased by 4.0 percentage points. While the baseline randomization inference yields conservative p-values ($p = 0.07$ and $p = 0.34$, respectively), this imprecision stems from the unweighted difference-in-differences design. By comparing the treated program against a heterogeneous set of control disciplines, the baseline model introduces substantial residual variance that partially obscures the treatment signal. When I employ PSM and SDiD specifications, the statistical precision improves markedly. By reweighting the control group to minimize compositional differences and optimize pre-treatment fit, these methods effectively filter out noise from poorly matched control units, reducing the error variance. Consequently, both specifications yield statistically significant improvement in graduation estimates (PSM: 9.0 pp, $p = 0.027$; SDiD: 5.6 pp, $p = 0.033$). Ultimately, these methods resolve the trade-off between conservative inference and estimation precision by minimizing between-cluster variance, confirming that the baseline imprecision was driven by identifying variation rather than bias.

The SW deadline regime generated improvements similar in magnitude to the BA effects but with notably greater statistical uncertainty in the primary specification. In the unadjusted difference-in-differences specification, treated students earned roughly 7.8 additional credits by semester four and 12.4 additional credits by semester seven, representing gains of about 10% and 9% relative to pre-reform cohorts. Although the estimated effects move consistently in the expected direction, they do not reach statistical significance under conservative randomization-inference procedures (RI-t $p > 0.20$). This lack of significance reflects the limited precision inherent in a setting with only one treated program and demographic differences between the treated and control groups.

To improve comparability and mitigate statistical noise, I re-estimate the effects using PSM and SDiD specifications. Both methods yield convergent evidence of substantial effects: the matched specification estimates 10.1 additional credits by semester four ($p = 0.017$) and 17 credits by semester seven ($p = 0.015$). The synthetic DiD framework corroborates this momentum, yielding significant gains in long-term credits (11.5 credits, $p = 0.002$) and a 7.1 percentage-point increase in nine-semester completion ($p = 0.010$). The close alignment of point estimates across specifications suggests that baseline imprecision reflected power constraints rather than a null response, though conclusions for Social Work require greater caution than for BA, given the heavier reliance on modeling assumptions.

Evidence on heterogeneous treatment effects indicates that the benefits of deadlines are broadly shared rather than concentrated among specific ability groups. In both the BA and SW programs, triple-difference tests reveal no statistically significant heterogeneity with respect to pre-university GPA ($p > 0.20$), implying that students of all ability levels benefit equally. While estimates for the lowest tercile are sometimes less precise due to statistical power, the consistent positive magnitudes across groups suggest that deadlines function as a universal pacing tool.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the German higher education system and the institutional setting of the university under study. Section 4 details the 2016 policy reforms. Section 5 presents administrative data and descriptive evidence. Section 6 explains the empirical strategy, including the inference procedures. Section 7 reports the main results, covering short- and long-run effects, event-study evidence, and heterogeneous treatment effects by student ability. Section 8 provides robustness checks using propensity-score–matched difference-in-differences estimators. Section 9 re-estimates effects using synthetic difference-in-differences to optimize pre-treatment fit. Section 10 discusses the findings and concludes.

2. Literature Review

***Deadlines as Incentives*—**Students often delay costly effort because the benefits of studying lie in the future while the costs are immediate, creating predictable time-inconsistency and procrastination (O'Donoghue & Rabin, 1999; Akerlof, 1991). In environments with weak external constraints, this mismatch leads to underinvestment in effort. Deadlines can mitigate these failures by raising the short-run cost of delay and helping individuals commit to effort schedules they struggle to sustain voluntarily (Ariely & Wertenbroch, 2002). Theoretically, these constraints function through several psychological channels: they heighten loss aversion by creating salient reference points (Zamir, Lewinsohn-Zamir & Ritov, 2017), and they reduce the psychological distance to future costs, thereby generating goal-gradient responses where effort intensifies as the deadline approaches.

Empirical work confirms that deadlines can increase timely effort. Ariely and Wertenbroch (2002) show that externally imposed, evenly spaced deadlines improve performance relative to flexible schedules or self-imposed deadlines. Evidence outside education shows similar dynamics: workers voluntarily accept penalty-based deadlines that concentrate effort (Kaur, Kremer & Mullainathan, 2015).

In the German higher-education system, Ostermaier (2018) shows that removing a binding deadline for introductory modules led students to strategically postpone effort. When the deadline was abolished, students increasingly submitted blank exams, thereby extending their study duration. The response was most pronounced among mid-ability students, illustrating that even modest institutional structure can discipline time-inconsistent study behavior in highly flexible environments. However, design matters: excessively rigid or poorly timed deadlines can reduce completion (Bisin & Hyndman, 2020). This paper examines institutional course-completion deadlines, a relatively unexplored policy tool distinct from assignment deadlines, and tests whether early versus delayed designs generate different behavioral responses.

Academic Dismissal Policies: Incentives versus Selection—A second related strand of research examines academic dismissal policies (AD), which set minimum first-year performance thresholds that students must meet to remain enrolled. A central question in this literature is whether such rules primarily change behavior (incentives) or merely alter which students persist (selection). The evidence is mixed, even within the Dutch higher-education context, where most studies are conducted.

Several studies conclude that AD policies function primarily as selection mechanisms. Arnold (2015) reports that while credit thresholds increased early attrition by 6–7 percentage points and accelerated the progress of survivors, they left the four-year completion rate for the starting cohort unchanged. This indicates that in some settings, dismissal simply brings dropout forward in time without generating net gains in human capital. Cornelisz et al. (2020) provide the most direct evidence of this selection channel: using a regression discontinuity design, they show that students just above and below the dismissal cutoff exhibit identical long-run graduation rates, implying that the dismissal threat itself did not causally alter academic trajectories. Sneyers and De Witte (2017) offer a slightly more positive view on efficiency, finding that while AD policies increased first-year dropout by roughly 7.5 percentage points, they also led to higher graduation rates among reenrolled students, suggesting that rigorous selection can improve success rates for those who persist.

Other studies examining comparable Dutch policies reach opposite conclusions. Schmidt et al. (2022) find that first-year credit requirements increased first-year performance without raising dropout, clear evidence of substantial effort responses. Effects concentrated among the weakest students, with completion rates for the lowest-GPA group rising from 17% to 48%. Arnold (2023) similarly shows that tightening performance standards reduces procrastination and increases credit accumulation among low-prepared students. These findings suggest AD rules can induce genuine behavioral responses within the same institutional environment where other studies find only selection. This paper extends this literature by providing causal evidence from the German higher education system, analyzing whether targeted, early milestones can trigger similar incentive effects without the broad-scope intensity of full-curriculum dismissal policies.

Academic Momentum and Long-Run Impacts—The long-term effects examined in this study relate to the theory of academic momentum, which posits that early academic success generates self-reinforcing trajectories toward degree completion (Attewell et al., 2012). Empirical research shows that intensive early engagement—such as enrolling in a full course load during the first semester—raises graduation probabilities by 6 to 10 percentage points (Belfield et al., 2016; Attewell & Monaghan, 2016), whereas early disengagement, such as part-time enrollment in the first semester, reduces completion rates by 5 to 13 percentage points (Attewell et al., 2012). These findings suggest that early achievement builds academic human capital, strengthens study habits, and reinforces institutional commitment. Crucially, recent experimental evidence supports this causal channel; Angrist et al. (2022) find that financial aid interventions increase graduation rates primarily by inducing students to accumulate more credits during their first year.

However, while financial incentives can generate momentum, most existing evidence on structural pacing relies on observational data or student-driven choices. It remains unclear whether requiring students to adopt a faster pace generates the same benefits as voluntarily choosing one. This study addresses that gap by testing whether imposing external deadlines can trigger this positive feedback loop, examining if short-term compliance translates into durable gains in credit accumulation, persistence, and on-time graduation.

3. Institutional Setting

3.1 The German Higher Education System

Germany's higher education system has two main types of institutions: research-oriented universities and practice-focused universities of applied sciences (UAS). Together, they enroll around 2.9 million students, with UAS institutions accounting for approximately 38% of all bachelor's enrollments (Statistisches Bundesamt, 2024). UAS institutions form a core component of the German tertiary system, providing practice-oriented education that integrates theoretical training with professional application. German universities organize their programs under the European Credit Transfer and Accumulation System (ECTS), which standardizes academic progression across Europe. This framework outlines a clear path to timely graduation, recommending that full-time bachelor's students complete 30 credits per semester.

In reality, few students adhere to this pace. Among graduates nationwide in 2023, the average time to degree was 8.5 semesters. Only 29.8% graduated on time, while 41.8% needed up to an extra year, and 28.4% took three or more additional semesters (Statistisches Bundesamt, 2024). Beyond these delays, dropout rates are high: approximately 30% of students abandon their studies, typically after 4.7 semesters (Heublein et al., 2017). These patterns reveal substantial inefficiencies in academic progression and the student-program matching process.

Three institutional characteristics contribute to these inefficiencies. First, tuition fees are mostly subsidized, which reduces the financial burden of prolonged enrollment[1]. Second, academic culture places strong emphasis on student autonomy and self-regulation, with few binding deadlines (Bäulke & Dresel, 2023)—a lack of structure that has been shown to inhibit academic progress (Scott-Clayton, 2015). Third, examination policies allow students to defer or repeat exams multiple times, which enables the strategic postponement of effort and extends study duration (Ostermaier, 2018). Together, these features create an environment where delaying progress carries little immediate cost, making procrastination particularly likely (O'Donoghue & Rabin, 1999).

3.2 The Institutional Context

The policy reform analyzed in this paper was implemented at a large UAS in Germany. The studied university enrolls roughly 12,000 students across more than twenty bachelor's programs. Its structure is centrally administered but with program-level autonomy over examination and progression policies, which creates within-institution variation ideal for identifying how academic policies influence student outcomes.

***Study Programs*—**The analysis focuses on two large programs: BA and SW, which together account for 18% of university enrollment. Both are nationally significant. Business Administration is one of Germany's most popular fields, accounting for 8.5% of first-year students, while Social Work accounts for 4.1% of total enrollment (121,664 students nationwide in 2024/2025). Both programs belong to the law, economics, and social sciences category, the largest field group in

[1] Public universities charge only modest administrative fees (typically around €250 per semester), and enrollment provides students with subsidized housing, healthcare, and transportation, potentially incentivizing prolonged study.

German higher education, enrolling 37.9% of all students (Statistisches Bundesamt, 2024). Both programs follow a structured seven-semester curriculum consistent with the ECTS framework, including a mandatory practical semester to link theory with workplace experience. While these programs had pre-existing academic progression rules (see section 4), the 2016 reform introduced significantly stricter deadlines, sharply reducing student flexibility.

4. The Policy Reform

In 2016, two of the university's largest bachelor's programs, BA and SW, introduced course-completion deadlines through changes to their study and examination regulations. The changes were implemented quietly through unannounced regulatory amendments[2], minimizing any potential selection into or out of the affected programs. It is unlikely that prospective students adjusted their choices in response to the new deadlines, as detecting these changes would have required a detailed examination of program regulatory documents, a rare practice among prospective students. The following subsections provide a detailed description of each reform, with Table 1 summarizing the policy parameters.

4.1 Business Administration: Early, Targeted Deadline

Pre-2016 Policy—Before April 2016, BA students faced a single deadline: Introduction to Business had to be completed by the end of the second semester. Students were allowed two examination attempts and were automatically re-registered for the next available session following failure. Exhausting both attempts resulted in academic dismissal.

Post-2016 Policy—The April 2016 reform significantly intensified this regime, shifting toward a high-stakes, early-intervention model. This intensification occurred along two dimensions. First, the deadline was advanced from the end of the second semester to the end of the first (temporal intensification). Second, the policy's scope was expanded to include "Business Mathematics" as a second mandatory course with a deadline (scope expansion). Consequently, students were required to complete both "Introduction to Business" and "Business Mathematics" by the end of their first semester. Failure to do so triggered automatic re-registration for the second semester. Failure to

[2] Aside from the introduction of binding course-completion deadlines, the 2016 regulatory updates did not alter the first-year curriculum, course content, examination rules, or degree requirements. All other changes consisted of minor editorial or administrative adjustments—such as updated module titles and coding, that left academic content and workload unchanged (see Appendix Table A6 for a summary).

pass by the second attempt (end of year one) resulted in academic dismissal[3]. The BA reform, therefore, represents an early, highly stringent, and targeted enforcement regime.

4.2 Social Work: Comprehensive, Broad Deadline

***Pre-2016 Policy*—**Before July 2016, SW students studied under a flexible system with no course-completion deadlines. Examinations could be scheduled or deferred freely across semesters. Students were allowed three exam attempts per module, with no risk of dismissal except after final failure.

***Post-2016 Policy*—**The July 2016 reform introduced deadlines, requiring the completion of all first-year modules (60 credits) by the end of Semester 4. Examinations not completed by this deadline were automatically recorded as failed, counting as one of the three permitted attempts. Missed or failed exams were automatically re-registered for the next available session, and students who exhausted all three attempts in any module were dismissed from the program. Consequently, a student who fails to meet the initial deadline and subsequently fails the two automatic re-sits would face academic dismissal at the end of the sixth semester.

This moderate-intensity, delayed-intervention policy was designed to promote steady academic progress by introducing time-bound accountability while giving students sufficient flexibility to adjust during their first two years. In practice, the combination of the three-attempt rule and the annual scheduling of many modules meant that students could remain enrolled well beyond the nominal deadline. Because failed exams triggered automatic re-registration for the next available session, which sometimes did not occur until the following academic year, the dismissal process could stretch over several additional semesters.

4.3 Comparing the Two Policies

The BA and SW reforms differ along three central dimensions. Timing: BA deadlines bind in the first semester, whereas SW deadlines apply at the end of the fourth. Scope: BA targets two foundational orientation modules, while SW covers all first-year coursework. Intensity: BA operates under a two-attempt rule with immediate consequences, whereas SW provides three

[3] These two modules are designated as 'orientation modules' (*Orientierungsprüfung*), which carry stricter conditions than the rest of the curriculum. While these specific courses are limited to one retake (two attempts total) and subject to automatic re-registration, standard modules in the program allow for three examination attempts, do not trigger automatic re-registration, and can be deferred by the student.

attempts and a longer compliance window. These design differences produce contrasting treatments—an early, targeted, high-stakes intervention in BA versus a broader, delayed, moderate-stakes intervention in SW. Evaluating each program separately enables a direct assessment of how the timing and intensity of deadlines shape student responses.

Table 1: Overview of Deadline Policies in Business Administration and Social Work

Policy Dimension	Business Administration (BA)		Social Work (SW)	
	Pre-2016	Post-2016	Pre-2016	Post-2016
Deadline Timing	Semester 2	Semester 1	None	Semester 4
Modules Covered	Intro to Business	Intro to Business; Business Math	No mandatory requirements	All first-year modules
Examination Attempts	2 attempts	2 attempts	3 attempts	3 attempts
Enforcement Mechanism	Automatic re-registration	Automatic re-registration	Flexible retake policy	Automatic re-registration
Implied Dismissal	Semester 3	Semester 2	N/A	Semester 6

Notes: "Automatic re-registration" means that students who fail or miss an exam are automatically enrolled in the next available examination attempt. Academic dismissal occurs upon exhausting the maximum number of allowed attempts in any mandatory module. "None" indicates the absence of binding course-completion deadlines before 2016.

5. Data and Descriptives

5.1 Data Sources and Structure

The analysis uses administrative student-level data from the university's examination office, covering all bachelor's programs from 2012 to 2022. The data are structured as a student-semester panel, tracking individual academic trajectories from enrollment through graduation, exit, or the end of the observation window. For the main analysis, the sample includes cohorts entering between 2012 and 2018[4]. This window allows tracking each student for at least four years, up to 2022, ensuring sufficient observation for long-term outcomes such as graduation. Cohorts entering between 2012 and 2015 constitute the pre-reform (control) period, while those starting from 2016 onward define the post-reform (treatment) period.

[4] The 2012–2018 range was selected to maximize comparability while avoiding confounding institutional or curricular changes that occurred before 2012 or after 2018.

The difference-in-differences design compares outcomes in two treated programs (BA & SW) to outcomes in 14 untreated study programs at the same university. The comparison group consists predominantly of STEM programs that experienced no deadline policy changes between 2012 and 2018. While these programs span different faculties, they share the same institutional environment and face common external factors affecting students, including macroeconomic conditions, and university-wide policies. This common framework supports the validity of the comparison group, provided pre-reform trends are parallel, a condition I test via event studies.

5.2 Outcome Variables

I examine treatment effects on both short-term and long-term academic outcomes to assess immediate behavioral responses at the point when deadlines bind and sustained momentum effects. The analysis also utilizes student background characteristics from university administrative records, including age, gender, type of high school degree, years since high school graduation, high school GPA, and citizenship status. These pre-determined covariates serve as control variables in the difference-in-differences specifications to improve precision and assess compositional trends.

Short-term outcomes are measured at the point when deadlines first bind: semester 2 for Business Administration and semester 4 for Social Work. These outcomes capture initial compliance and immediate adjustments to the new policy environment. I examine: (i) cumulative ECTS credits earned, the primary measure of academic progress; (ii) grade point average (GPA), capturing academic performance quality; (iii) cumulative dropout, defined as a binary indicator equal to one if the student exits the program without completing a degree; and (iv) early dropout, a dummy variable taking the value of one if the student drops out within the first academic year.

Long-term outcomes are measured at semester 7, the nominal program duration, to assess whether early gains translate into improved degree completion. I analyze: (i) cumulative ECTS credits, measuring sustained academic progress; (ii) GPA, testing whether quality deteriorates as students accelerate; (iii) cumulative dropout, defined as a binary indicator for students who exit the program, capturing persistence through the standard program duration; (iv) on-time graduation, a dummy variable equal to one if the student completes their degree within seven semesters; and (v) graduation within nine semesters, an indicator allowing one year beyond the nominal duration.

Graduation outcomes are analyzed for cohorts with sufficient follow-up time, excluding the 2018 cohort from the nine-semester completion analysis.

5.3 Descriptive statistics

A key identification requirement for the difference-in-differences design is that the 2016 reforms did not trigger compositional changes in the treated programs. Table 2 compares pre-reform (2012–2015) and post-reform (2016–2018) cohorts in BA, SW, and the comparison programs, reporting average characteristics and DiD estimates.

Measures of academic preparedness show no differential changes. DiD estimates for high school GPA and the share of students holding the Abitur are statistically insignificant and close to zero, suggesting that the reform did not induce selection on ability. Demographic characteristics also align with overall university trends and experience minimal differential movements.

One exception is the decline in female enrollment in BA (DiD=-6.65 pp, $p<0.01$), indicating a differential shift. This reflects divergent trends across fields: female representation decreased modestly in BA (58.8% to 55.0%) while increasing in comparison programs, which are predominantly STEM fields (24.9% to 27.9%). These opposing movements align with national patterns of declining female participation in business education alongside rising representation in STEM disciplines.

Crucially, this gender compositional shift likely yields conservative estimates. Since female students generally achieve higher outcomes in higher education (Verbree et al., 2023) and lower dropout rates (Heublein et al., 2017), a reduction in their share mechanically dampens average performance. Additionally, the SW program shows a small but statistically significant increase in the share of first-time university entrants (DiD = 3.9 pp). Overall, the stability of the key ability measures, high-school GPA, and Abitur share, supports the validity of the identification strategy, while the modest shifts in other characteristics mirror broader university-wide and national trends rather than responses to the reform.

Table 2: Descriptive Statistics and Trend in Students' Characteristics

	Pre-2016 (1)	Post-2016 (2)	Control Pre-2016 (3)	Control Post-2016 (4)	DiD (5)
Panel A: BA					
High School GPA	2.532	2.446	2.776	2.705	-0.01
					(0.02)
Abitur (%)	32.66	41.51	37.37	43.23	3.0
					(2.22)
Female (%)	58.77	55.01	24.95	27.85	-6.65***
					(2.226)
Age at Entry	21.67	21.59	21.65	21.65	-0.073
					(0.142)
German (%)	96.92	96.42	97.16	95.26	1.397*
					(0.842)
First Uni Semester (%)	0.839	0.764	0.824	0.734	0.014
					(0.018)
N	1,528	978	5,378	4,094	11,978
Panel B: SW					
High School GPA	2.388	2.279	2.776	2.705	-0.03
					(0.02)
Abitur (%)	39.32	45.44	37.37	43.23	0.27
					(2.45)
Female (%)	81.58	83.16	24.95	27.85	-1.310
					(1.942)
Age at Entry	23.27	23.46	21.65	21.65	0.189
					(0.265)
German (%)	97.29	96.96	97.16	95.26	1.571*
					(0.859)
First Uni Semester (%)	0.830	0.780	0.824	0.734	0.039**
					(0.020)
N	1,292	790	5,378	4,094	11,554

Notes: This table reports mean pre- and post-reform characteristics of students in treated and control programs for Business Administration (Panel A) and Social Work (Panel B). Columns (1)–(4) report group means before and after the 2016 reform. Column 5 presents the difference-in-differences estimate for each characteristic. Standard errors are reported in parentheses. High School GPA is the official secondary-school leaving grade (lower values indicate better performance). Abitur indicates possession of the academic-track university entrance certificate. Female and German refer to gender and citizenship status. *Age at Entry* is measured at first matriculation. First Uni Semester identifies first-time entrants.

6. Empirical Strategy

6.1 Difference-in-Differences Design

I estimate the causal effect of course-completion deadlines using a two-way fixed effects DiD design that compares outcome changes in treated programs (BA and SW) before and after 2016 with contemporaneous changes in 14 untreated programs at the same university. Program fixed effects absorb time-invariant differences across disciplines, such as grading cultures or inherent curriculum difficulty, while cohort fixed effects capture shocks common to each entering cohort, such as changes in macroeconomic conditions or university-wide administrative policies.

A simple pre-post comparison of treated students risks confounding deadline effects with time trends, while a cross-sectional comparison in 2016 conflates treatment effects with inherent program differences. The Difference-in-Differences (DiD) design eliminates both biases by comparing temporal changes in treated programs against those in control programs. The following specification is estimated separately for BA and SW:

$$Y_{ipt} = \alpha + \beta \left(Treated_p \times Post_t\right) + \gamma' X_i + \mu_p + \lambda_t + \varepsilon_{ipt} , \qquad (1)$$

Where Y_{ipt} is the outcome of interest for student i in program p from entry cohort t. $Treated_p$ is an indicator equal to one for the treated program (e.g., BA) and zero for comparison programs. $Post_t$ is an indicator equal to one for cohorts enrolling in or after 2016. The vector X_i includes pre-determined student characteristics[5]: high school GPA, Abitur holder, gender, age at immatriculation, German citizenship status, and an indicator for being in the first university semester. μ_p and λ_t represent program and cohort fixed effects, respectively. Standard errors are clustered at the program level to account for within-program correlation.

The coefficient β identifies the average treatment effect on the treated under the parallel-trends assumption. This assumption requires that, absent the reform, treated and control programs would have followed similar outcome trajectories. I assess the validity of this assumption using event studies (see section 6.2). Finally, I complement the primary analysis with matched DiD (Section

[5] Zeldow and Hatfield (2021) illustrate that time-invariant covariates will not introduce bias in DiD analysis if their effect on the outcome is constant over time. Following this, time-invariant covariates are included in the model to improve precision. These covariates show similar trends across groups, and their effect on student outcomes remains stable across semesters.

8) and Synthetic Difference-in-Differences (Section 9). These methods strengthen causal identification by restricting comparisons to observably similar students and constructing weighted counterfactuals that mirror the treated program's pre-reform trajectory with greater precision than standard linear DiD.

Inference with Few Treated Clusters—A central challenge for inference is the small number of clusters: the analysis includes only 15 study programs, and each reform applies to a single treated program. Because the estimation clusters standard errors at the study-program level, conventional cluster-robust variance estimators are prone to over-rejection in settings with few clusters (Cameron and Miller, 2015). To ensure valid inference, I rely primarily on randomization inference (RI-t) following MacKinnon and Webb (2020). By randomly permuting treatment assignment across programs (999 replications), this method generates exact finite-sample p-values independent of asymptotic assumptions. For transparency and completeness, I also report wild cluster bootstrap p-values (Cameron, Gelbach, and Miller, 2008) and conventional program-clustered standard errors.

6.2 Event-Study Specification

To assess the validity of the parallel-trends assumption and to trace the dynamic effects of the deadline policy across cohorts, I estimate an event-study version of the DiD model. The event-study replaces the single DiD interaction with a series of relative-cohort indicators that trace the evolution of outcomes before and after treatment adoption.

The event-study model is specified as follows:

$$Y_{ipt} = \alpha + \sum_{k=-4}^{-2} \delta_k \times Treated_{ptk} + \sum_{k=0}^{2} \delta_k \times Treated_{ptk} + \gamma' X_i + \mu_p + \lambda_t + \varepsilon_{ipt}\ , (2)$$

where k indexes cohort years relative to the policy implementation year (2016 = 0). $Treated_{ptk}$ is an indicator variable equal to one if treated program p in cohort t is observed k years relative to policy implementation, and zero otherwise. The first summation captures the pre-treatment coefficients for cohorts 2012 (k =- 4) through 2014 (k = -2). The 2015 cohort (k=-1) serves as the omitted reference category. The second summation captures post-treatment effects for cohorts 2016 (k=0) through 2018 (k=2).

The pre-treatment coefficients δ_k for k<0 test whether the treated and control programs exhibited differential trends before the policy. Under the parallel trends assumption, these coefficients should be jointly insignificant and close to zero. I formally test this using an F-test of joint significance and present the event-study estimates both graphically and in tabular form.

6.3 Heterogeneous Effects by Student Ability

Deadline policies may affect students differently depending on their prior academic preparation. To examine whether the reform generates heterogeneous responses across ability levels, I classify students into sample-wide terciles based on their high-school GPA (Low, Medium, High ability). I then estimate the following fully interacted triple-difference (DDD) specification:

$$Y_{ipt} = \alpha + \beta_M(Treated_p \times Post_t) + \beta_L(Treated_p \times Post_t \times Low_i) + \beta_H(BW_p \times Post_t \times High_i) + \alpha_L Low_i + \alpha_H High_i + \gamma' X_i + \mu_p + \lambda_t + \epsilon_{ipt}, \quad (3)$$

where β_L and β_H capture the differential treatment effects for low- and high-ability students relative to the medium-ability baseline β_M. Low_i and $High_i$ indicate the lowest and highest ability terciles, and the model includes the same program and cohort fixed effects and student-level controls as in the baseline DiD specification. This fully interacted triple-difference structure allows the treatment effect of the deadline policy to vary flexibly across ability groups. In the results section, I report the subgroup-specific treatment effects, obtained by re-estimating the baseline specification within ability groups, and directly present the estimated policy impact for low-, medium-, and high-ability students.

The DDD estimator requires parallel trends in ability gaps, i.e., that differences between ability groups would have evolved similarly in treated and untreated programs in the absence of the reform. I assess this assumption using an ability-specific event-study design and a joint pre-trend test of all pre-treatment interactions. These tests serve as the basis for the formal validation of the heterogeneous treatment effects (see Appendix Figures A1 and A2).

7. Results

This section presents the main empirical results in three parts. Sections 7.1–7.3 analyze the BA reform, reporting short-term effects (Section 7.1), long-term effects (Section 7.2), and event-study evidence supporting parallel pre-trends (Section 7.3). Sections 7.4–7.6 present parallel analyses for SW. Section 7.7 examines whether effects vary by student ability, testing whether deadlines disproportionately harm weaker students.

7.1 Business Administration: Short-Term Effects

The introduction of first-semester deadlines led to a substantial, immediate increase in academic effort. Table 3 reports the short-run DiD effects measured at the end of Semester 2. By the end of the first year, treated students accumulated 7.15 additional ECTS credits (SE = 0.83), an 18% increase over the pre-treatment mean of 40.3 credits. This effect is highly statistically significant (RI-t p = 0.001) and equivalent to approximately 1.4 additional courses. Event-study estimates (Figure 1) confirm that these gains emerged sharply following the 2016 reform, with no evidence of differential pre-trends.

Table 3: Short-Term Effects of Course-Completion Deadline Policy (Semester 2): BA

	Credits (1)	GPA (2)	Dropout (3)	Early Dropout (4)
DiD Estimate	7.15***	-0.01	-0.01	0.10
	(0.83)	(0.02)	(0.01)	(0.01)
RI-t p-value	[0.001]	[0.93]	[0.92]	[0.21]
Bootstrap p-value	[0.06]	[0.58]	[0.56]	[0.18]
Pre-treatment Mean	40.27	2.57	0.13	0.522
Effect Size (%)	17.8%	-0.7%	-8.6%	19.1%
Controls	Yes	Yes	Yes	Yes
Cohort & Program FE	Yes	Yes	Yes	Yes
Observations	11,877	11,877	11,877	5,779

Notes: The table reports DiD estimates for BA, obtained by OLS, for outcomes measured at the end of Semester 2. All models include individual controls (high school GPA, Abitur type, gender, age, first-university-semester indicator, foreign citizenship), cohort fixed effects, and study-program fixed effects. Standard errors clustered at the program level (15 clusters) are reported in parentheses. Wild-cluster bootstrap p-values and RI-t p-values from randomization inference appear in brackets. Effect sizes are calculated relative to the pre-treatment mean. "Early Dropout" denotes the share of dropouts during the first academic year. Significance stars are based on RI-t p-values: *** p<0.01, ** p<0.05, * p<0.10 .

Importantly, this credit gain occurred without compromising academic quality or triggering mass dropout. The estimated effect on GPA is negligible and statistically insignificant (coefficient = -

0.01, RI-t p = 0.93), indicating that students expanded total study effort rather than reallocating a fixed amount of effort across a larger course load. Similarly, the reform did not increase overall attrition. Cumulative dropout by Semester 2 remains unchanged (–0.01, RI-t p = 0.92), suggesting that the additional academic pressure induced by the earlier deadline did not trigger excess exits from the program.

Finally, I examine whether the policy affected the timing of dropout. Early dropout among eventual leavers increased by 10 percentage points, though this estimate is imprecise in the baseline specification (RI-t p = 0.21). Propensity-score matching substantially improves precision and confirms a significant effect (Section 8.1), suggesting the reform accelerated exit timing among students who would have dropped out rather than affecting overall dropout levels.

7.2 Business Administration: Long-Term Effects

Early gains in credit accumulation translated into sustained momentum rather than a one-time intertemporal substitution of effort. Table 4 reports the effects on long-term progression and degree completion. Treated students accumulated 21.9 additional credits by Semester 7 (SE = 3.14, RI-t p = 0.054), a 17% gain relative to the pre-treatment mean. Notably, this proportional increase mirrors the 18% gain observed at Semester 2, demonstrating that the early acceleration initiated a self-reinforcing trajectory rather than merely shifting effort forward in time. Importantly, this credit gain is accompanied by continued stability in overall grades, as cumulative GPA remained unchanged (RI-t p = 0.81).

These sustained credit gains translated into higher degree completion rates, though here the evidence must be interpreted with greater caution. Graduation within nine semesters increased by 7.0 percentage points (SE = 0.011, RI-t p = 0.07), a 12% rise relative to the pre-reform mean. While the on-time graduation increases by 4.0 percentage points (SE = 0.01), the effect is less precisely determined (RI-t p = 0.34). However, the results prove robust across alternative identification strategies; both the matched DiD and synthetic DiD frameworks yield highly significant point estimates for both on-time ($p < 0.001$) and nine-semester graduation (p = 0.033). (see sections 8.1 and 9.1).

Lastly, the results contradict the hypothesis that strict deadlines improve averages solely by selecting out weaker students. Cumulative dropout declined by 5.5 percentage points. While this estimate is imprecise under conservative inference (RI-p = 0.33), it yields a statistically significant reduction in the propensity-score matched analysis (see Section 8.1). This indication of improved retention shows that binding course deadlines foster persistence rather than inducing exit. Moreover, by forcing early engagement, the policy strengthened the institutional commitment and academic habits necessary for persistence, as outlined by the academic-momentum framework (Attewell et al., 2012).

Table 4: Long-Term Effects of Course-Completion Deadline Policy: Business Administration

	Credits (1)	GPA (2)	Dropout (3)	On-Time Grad. (4)	Grad. ≤ 9 Sem. (5)
DiD Estimate	21.86*	-0.01	-0.05	0.04	0.07*
	(3.14)	(0.01)	(0.01)	(0.01)	(0.01)
RI-t p-value	[0.05]	[0.81]	[0.33]	[0.34]	[0.07]
Bootstrap p-value	[0.08]	[0.58]	[0.25]	[0.22]	[0.08]
Pre-treatment Mean	130.48	2.43	0.26	0.096	0.571
Effect Size (%)	16.8%	-0.5%	-21.2%	41.5%	12.2%
Controls	Yes	Yes	Yes	Yes	Yes
Cohort & Program FE	Yes	Yes	Yes	Yes	Yes
Observations	11,877	11,877	11,877	11,877	10,211

Notes: The table reports DiD estimates for Business Administration, obtained by OLS, for outcomes measured at Semester 7 and for graduation outcomes. All specifications include individual controls (high school GPA, Abitur type, gender, age, first-university-semester indicator, foreign citizenship) as well as cohort and study-program fixed effects. Standard errors clustered at the program level (15 clusters) are shown in parentheses. Wild-cluster bootstrap p-values and RI-t p-values based on randomization inference are reported in brackets. Effect sizes are computed relative to the pre-treatment mean. "On-Time Graduation" denotes completing the degree in Semester 7; "Graduated ≤ 9 Semesters" excludes the 2018 cohort. Significance stars are based on RI-t p-values: *** p<0.01, ** p<0.05, * p<0.10.

7.3 Business Administration: Event Study and Sensitivity Analysis

Figure 1 presents the event-study estimates for Business Administration, plotting treatment effects relative to the 2015 pre-reform baseline. Pre-treatment coefficients fluctuate around zero with no statistical evidence of systematic drift, despite some visual dispersion reflecting inherent sampling variation (particularly in panel B and D). Formal bootstrap joint tests fail to reject parallel trends for all key outcomes: Semester 2 credits (p = 0.51), Semester 7 credits (p = 0.31), on-time graduation (p = 0.41), and nine-semester graduation (p = 0.32) (see Appendix Table A1). This provides statistical support for the parallel pre-treatment trends assumption underlying the DiD design.

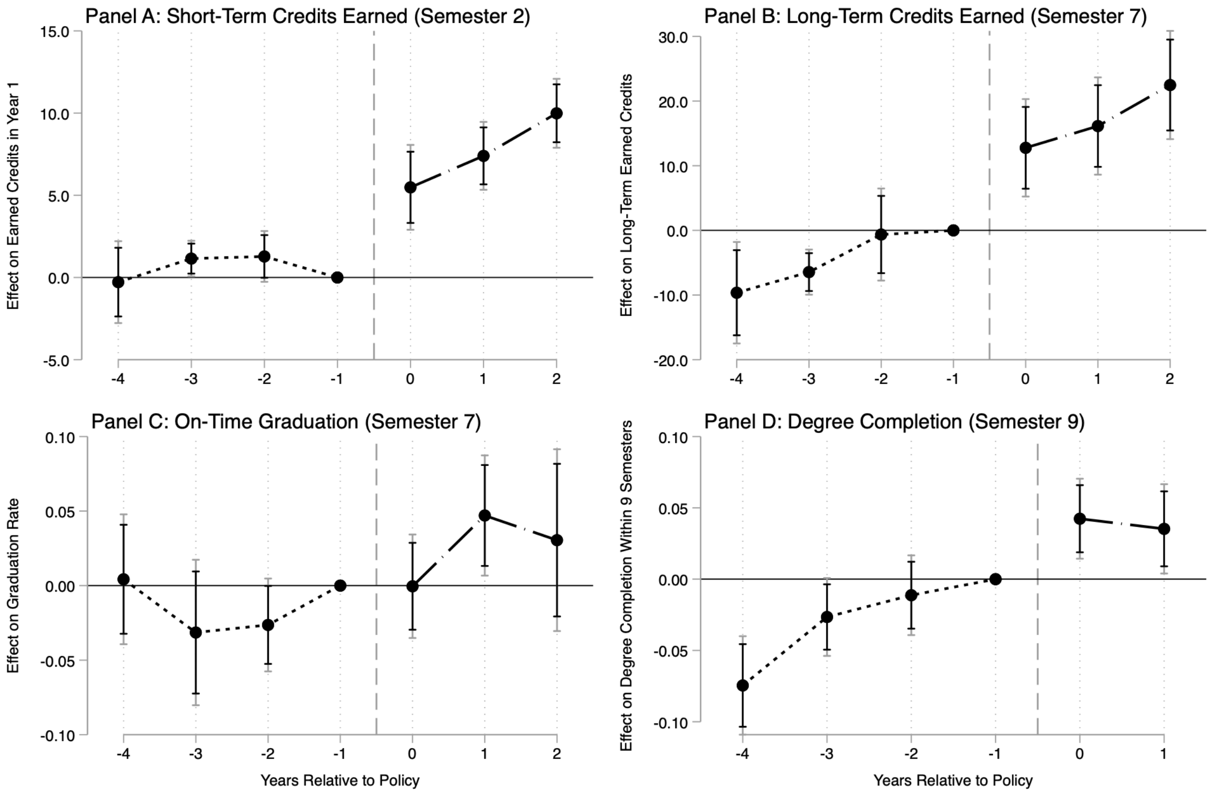

Figure 1: Event Study Graphs: Business Administration

Notes: The figure plots event-study coefficients for Business Administration from specifications estimating dynamic treatment effects relative to the 2015 pre-reform cohort. Panels A and B report effects on credits earned in Semesters 2 and 7; Panels C and D report effects on on-time graduation (Semester 7) and graduation within nine semesters. All models include cohort and program fixed effects and standard student-level controls. Point estimates are shown with 95% (grey) and 90% (black) confidence intervals based on standard errors clustered at the program level. The vertical dashed line marks the first treated cohort (2016). Full regression output is provided in Appendix Table A1.

Sensitivity Analysis: Parallel Trends—To assess the validity of the identification strategy, I implement the sensitivity analysis proposed by Rambachan and Roth (2023). This method constructs robust confidence intervals by allowing for deviations from exact parallel trends, parameterized by $\bar{M}$ times the maximal observed pre-treatment trend difference.

Appendix Table B1 reports the results. The impact on credit accumulation proves robust to substantial trend violations. The effect on short-term credits (Semester 2) remains statistically significant even if post-treatment trend deviations are 1.25 times larger than the maximum pre-treatment deviation ($\bar{M} = 1.25$). Long-term credit accumulation (Semester 7) also exhibits resilience ($\bar{M}$ between 0.75 and 1.25). In contrast, the effect on graduation within nine semesters is more sensitive; the estimate remains distinguishable from zero only if trend deviations do not exceed 25% of the pre-treatment maximum ($\bar{M} = 0.25$). These results suggest that while the reform drove a robust increase in credit earning, the downstream effect on final graduation rates relies more heavily on the assumption of trend stationarity.

This sensitivity highlights the importance of the Synthetic DiD and Matching specifications employed later in this paper, which explicitly reweight the control group to minimize such trend differentials. Collectively, the sensitivity analysis confirms that the reform drove a robust, causal acceleration in study progress, while the magnitude of the final graduation effect is best interpreted in conjunction with the comprehensive suite of robustness checks.

7.4 Social Work: Short-Term Effects

The introduction of the relatively lenient and comprehensive deadline in SW generated academic improvements similar to the BA reform, though estimated with less statistical precision. Table 5 reports the short-term effects in SW, measured at Semester 4 when the delayed deadline first binds. Treated students earned 7.8 additional credits by semester four (SE = 1.86), roughly a 10% increase over the pre-reform mean. While this point estimate is economically meaningful, it does not reach statistical significance under the conservative randomization inference used for the primary specification (RI-t $p = 0.21$).

This lack of significance reflects the conservative nature of randomization inference in a setting with a single treated cluster. With only 15 programs, the RI-t procedure correctly interprets the substantial between-cluster heterogeneity as noise, resulting in wider confidence intervals. Specifically, precision is constrained by three structural factors: (i) the single treated unit limits variation; (ii) Social Work cohorts are smaller than Business Administration cohorts; and (iii) Social Work students differ demographically from control programs. Consequently, even large point estimates rarely fall in the extreme tails of the placebo distribution, explaining the lack of significance despite favorable magnitudes.

Propensity-score matching directly addresses this compositional imbalance. As shown in Section 8.2, the matched difference-in-differences specification yields a 10.1-credit increase that is statistically significant (bootstrap $p = 0.01$), strongly suggesting that the lack of significance reflects power limitations rather than the absence of a behavioral response. Effects on other short-term outcomes are minimal. Dropout falls by 4.9 percentage points, but again the estimate is statistically indistinguishable from zero (RI-t $p = 0.45$). GPA worsens modestly, although this effect is neither significant nor persistent, as it fully dissipates by semester seven.

Overall, while the primary difference-in-differences provides directionally favorable evidence, the baseline estimates are too imprecise to reach conventional significance levels under conservative randomization inference. However, as shown in the matched specification (Section 8.2) and synthetic DiD (Section 9.2), addressing the compositional imbalance and optimizing the counterfactual significantly improves statistical precision. The close alignment of point estimates across these diverse specifications suggests that the initial lack of significance was driven by power constraints rather than a null response. Consequently, these refined models provide robust evidence that the Social Work reform initiated a genuine shift in academic trajectories.

A comparison across identical time horizons reveals that the early BA deadline triggers immediate, self-reinforcing academic momentum that grows significantly between semesters 2 and 4, whereas the SW program exhibits a more gradual anticipatory response before its primary binding point. As detailed in Appendix Table A3, the magnitude of credit accumulation in the BA program nearly doubles over this period, while the SW program requires the actual binding of the deadline to reach similar levels of effort. This divergence suggests that the timing of institutional constraints is a primary determinant of the speed at which students adjust their academic pace.

Table 5: Short-Term Effects of Course-Completion Deadline Policy (Semester 4): Social Work

	Credits (1)	GPA (2)	Dropout (3)	Early Dropout (4)
DiD Estimate	7.79	0.06	–0.04	–0.02
	(1.86)	(0.01)	(0.02)	(0.01)
RI-t p-value	[0.21]	[0.15]	[0.45]	[0.54]
Bootstrap p-value	[0.20]	[0.38]	[0.32]	[0.46]
Pre-treatment Mean	81.28	2.12	0.20	0.84
Effect Size (%)	9.6%	2.9%	–24.4%	–3.5%
Controls	Yes	Yes	Yes	Yes
Cohort & Program FE	Yes	Yes	Yes	Yes
Observations	11,454	11,454	11,454	5,516

Notes: The table reports DiD estimates for SW, obtained by OLS, for outcomes measured at the end of Semester 4. All models include individual controls (high school GPA, Abitur type, gender, age, first-university-semester indicator, foreign citizenship), cohort fixed effects, and study-program fixed effects. Standard errors clustered at the program level are reported in parentheses. Wild-cluster bootstrap p- and RI-t p-values from randomization inference appear in brackets. Effect sizes are calculated relative to the pre-treatment mean. "Early Dropout" denotes dropout during the first academic year. Significance stars are based on RI-t p-values: *** p<0.01, ** p<0.05, * p<0.10.

7.5 Long-Term Effects: Social Work

Long-term estimates for Social Work are consistent with sustained momentum, though they lack statistical precision in the primary specification (see Table 6). By Semester 7, treated students had accumulated an estimated 12.4 additional credits relative to the control group (SE = 3.23). This point estimate represents a 9% gain over the pre-reform mean, suggesting that students maintained the proportional advantage gained in the first two years rather than falling back to their pre-reform pace. However, under conservative randomization inference, this estimate does not reach statistical significance (RI-t p = 0.23).

This pattern extends to degree completion. The point estimate for on-time graduation indicates an increase of 4.6 percentage points (Column 4), representing a proportional rise of roughly 42% relative to the low pre-reform baseline. As with the short-term results, this estimate is directionally favorable but statistically imprecise (RI-t p = 0.29), reflecting the same power limitations and single-cluster design constraints discussed in Section 7.4.

Table 6: Long-Term Effects of Course-Completion Deadline Policy: Social Work

	Credits (1)	GPA (2)	Dropout (3)	On-Time Grad. (4)	Graduated ≤ 9 Semesters (5)
DiD Estimate	12.35	0.01	–0.03	0.04	0.01
	(3.23)	(0.01)	(0.01)	(0.01)	(0.01)
RI-t p-value	[0.23]	[0.86]	[0.78]	[0.29]	[0.51]
Bootstrap p-value	[0.23]	[0.62]	[0.37]	[0.24]	[0.43]
Pre-treatment Mean	142.64	2.04	0.22	0.11	0.59
Effect Size (%)	8.7%	0.6%	–14.1%	41.7%	3.3%
Controls	Yes	Yes	Yes	Yes	Yes
Cohort & Program FE	Yes	Yes	Yes	Yes	Yes
Observations	11,454	11,454	11,454	11,454	9,851

Notes: The table reports DiD estimates for Social Work, obtained by OLS, for outcomes measured at Semester 7 and for graduation outcomes. All specifications include individual controls (high school GPA, Abitur type, gender, age, first-university-semester indicator, foreign citizenship) as well as cohort and study-program fixed effects. Standard errors clustered at the program level (15 clusters) are shown in parentheses. Wild-cluster bootstrap p-values and RI-t p-values based on randomization inference are reported in brackets. Effect sizes are computed relative to the pre-treatment mean. "On-Time Graduation" denotes completing the degree in Semester 7; "Graduated ≤ 9 Semesters" excludes the 2018 cohort. Significance stars are based on RI-t p-values: *** p<0.01, ** p<0.05, * p<0.10.

Despite the lack of significance in the primary specification, four pieces of evidence suggest that these estimates reflect genuine behavioral responses rather than idiosyncratic noise or pre-existing trend differentials. First, event-study evidence (Section 7.6) identifies a structural break in trend precisely at the time of the reform, rather than a gradual drift. Second, the proportional persistence of the credit effect (10% in the short run, 9% in the long run) aligns with the theory of academic momentum; importantly, sensitivity analysis following Rambachan and Roth (2023) confirms these effects remain significant even under substantial extrapolation of potential pre-trends. Third, the matched specification (Section 8.2), which restricts comparisons to observably similar students, yields a significant increase in both graduation and credits earned. Fourth, the synthetic DiD analysis (Section 9.2) decisively resolves the statistical ambiguity of the baseline model by optimizing pre-treatment fit, corroborating that the observed gains are activated by the deadline rather than differential trends.

7.6 Event Study Evidence: Social Work

Figure 2 presents the event-study estimates for Social Work, plotting treatment effects relative to the 2015 pre-reform baseline. Pre-treatment coefficients display greater dispersion than in Business Administration, reflecting smaller cohort sizes, but exhibit no systematic trend. Formal bootstrap joint tests fail to reject parallel trends for all outcomes: Semester 4 credits ($p = 0.40$), Semester 7 credits ($p = 0.32$), on-time graduation ($p = 0.32$), and nine-semester graduation ($p = 0.33$) (see Appendix Table A2). However, acknowledging that these tests may be underpowered in a single-treated-unit setting, I further assess the results using sensitivity analysis following Rambachan and Roth (2023) (see Appendix B).

Sensitivity Analysis: Parallel Trends—Appendix Table B2 presents the sensitivity analysis results, revealing a distinct temporal heterogeneity in the reform's impact. For credit accumulation, the estimates for the initial treated cohorts (t=0, 1) are statistically significant but sensitive to trend assumptions (Breakdown $\bar{M} = 0.25$). However, the effect consolidates significantly for the third cohort (t=2), where the impact on short-term credits doubles in magnitude and achieves moderate robustness ($\bar{M} = 0.75$), while long-term credits become highly robust ($\bar{M} = 1.25$). This pattern suggests that while the immediate behavioral response in Social Work was noisy, the reform successfully established a robust, cumulative trajectory of increased study effort as the policy matured.

For graduation outcomes, the OLS baseline estimates are more fragile. While I observe marginally significant increases in on-time graduation for the first two cohorts, these results rely on strict trend stationarity ($\bar{M} = 0.25$) and do not persist for the third cohort. Furthermore, I find no significant baseline effect on graduation within nine semesters. The sensitivity of these OLS estimates to linear trend assumptions underscores the importance of the Synthetic DiD and Propensity Score Matching specifications presented earlier. By explicitly reweighting the control group to minimize pre-trend differentials, those methods provide stronger evidence for degree completion than the standard DiD specification, which appears sensitive to the higher trend variability observed in the Social Work sample. Thus, I interpret the sensitivity analysis not as evidence against the graduation effects, but as confirmation that proper control group construction is better achieved via the preferred SDID and matching specifications.

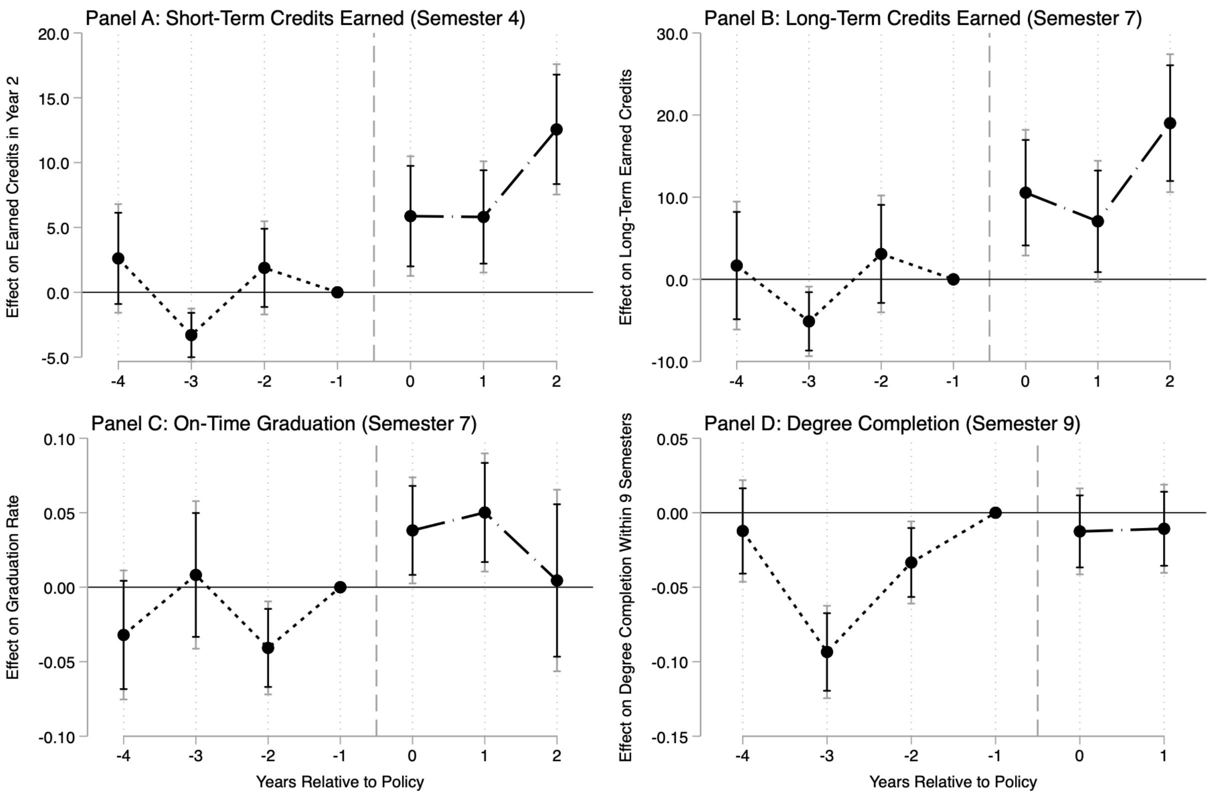

Figure 2: Event Study Graphs: Social Work

Notes: The figure plots event-study coefficients for Social Work from specifications estimating dynamic treatment effects relative to the 2015 pre-reform cohort. Panels A and B report effects on credits earned in Semesters 4 and 7; Panels C and D report effects on on-time graduation (Semester 7) and graduation within nine semesters. All models include cohort and program fixed effects and standard student-level controls. Point estimates are shown with 95% (grey) and 90% (black) confidence intervals based on standard errors clustered at the program level. The vertical dashed line marks the first treated cohort (2016). Full regression output is provided in Appendix Table A2.

7.7 Heterogeneous Treatment Effects by Student Ability

A central equity concern is whether binding deadlines help all students or disproportionately burden those entering university with weaker academic preparation. Deadlines could, in principle, function as gatekeeping mechanisms that screen out lower-ability students, or they may instead provide external structure that is especially valuable for those who struggle with self-regulation (Scott-Clayton, 2015).

To assess these competing hypotheses, I evaluate whether treatment effects differ across the ability distribution by grouping students into terciles of high-school GPA (High, Medium, Low) and estimating separate difference-in-differences models for each ability group (Tables 7 and 8). These are complemented by a fully interacted triple-difference specification that provides a joint test of equality across groups (Appendix Tables A4 and A5). Finally, event-study graphs assess whether pre-treatment ability gaps evolve differently across programs (Figures A1 and A2).

7.7.1 Heterogeneous Treatment Effects: Business Administration

The estimated treatment effects for Business Administration, presented in Table 7, provide evidence against the hypothesis that binding deadlines function as a gatekeeping mechanism. Instead, the data suggest a broad-based improvement in pacing with particularly striking relative gains for the lowest-performing students, although statistical precision varies across groups.

In the short run (Semester 2), high- and medium-ability students experience statistically significant increases in credit accumulation. High-ability students earn 8.34 additional credits (bootstrap $p = 0.01$), a 19 percent increase relative to their pre-treatment mean of 44.2 credits. Medium-ability students gain 6.42 credits (bootstrap $p = 0.02$), corresponding to a 17 percent increase over their baseline of 38.5 credits. Low-ability students exhibit the largest point estimate (11.21 credits), though imprecisely estimated (bootstrap $p = 0.23$). Nevertheless, this implies a substantial 33 percent increase relative to their lower baseline of 34.4 credits.

Long-run outcomes display a similar pattern. High-ability students accumulate 28.57 additional credits (bootstrap $p = 0.01$), a 20 percent increase relative to baseline, while medium-ability students gain 19.61 credits (bootstrap $p = 0.06$), a 16 percent increase. Low-ability students again show the largest point estimate—31.51 credits (bootstrap $p = 0.24$)—equivalent to a 28 percent increase relative to their baseline. Graduation within nine semesters rises by 9 percentage points for high-ability students (bootstrap $p = 0.07$) and 7 percentage points for medium-ability students (bootstrap $p = 0.09$), representing comparable proportional gains of roughly 13 percent. For low-ability students, the point estimate implies a 13-percentage-point increase (bootstrap $p = 0.20$), corresponding to a 30 percent relative improvement over a baseline graduation rate of 43 percent.

Table 7: Heterogeneous Treatment Effects by Student Ability: Business Administration

Panel A. Short-Term Effects (Semester 2)

	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
High Ability	8.34**	−0.05	−0.04**	0.01
	(0.97)	(0.03)	(0.01)	(0.02)
	[0.01]	[0.41]	[0.22]	[0.56]
Pre-treatment Mean	44.2	2.3	0.13	0.58
Observations	3,851	3,851	3,851	3,851
Medium Ability	6.42**	−0.02	−0.005	0.02
	(0.84)	(0.02)	(0.01)	(0.02)
	[0.02]	[0.42]	[0.75]	[0.51]
Pre-treatment Mean	38.5	2.7	0.13	0.52
Observations	3,881	3,881	3,881	3,881
Low Ability	11.21	0.01	−0.02	0.09
	(1.51)	(0.02)	(0.03)	(0.02)
	[0.23]	[0.66]	[0.50]	[0.41]
Pre-treatment Mean	34.4	2.8	0.13	0.41
Observations	4,127	4,127	4,127	4,127

Panel B. Long-Term Effects (Semester 7)

	Credit Points	GPA	Dropout	On-Time Graduation	Graduated ≤ 9 Semesters
High Ability	28.57**	−0.07	−0.10**	0.04	0.09*
	(3.01)	(0.02)	(0.01)	(0.02)	(0.01)
	[0.01]	[0.30]	[0.01]	[0.31]	[0.07]
Pre-treatment Mean	140.8	2.2	0.22	0.15	0.67
Observations	3,851	3,851	3,851	3,851	3,213
Medium Ability	19.61*	−0.00	−0.03	0.06*	0.07*
	(3.71)	(0.02)	(0.02)	(0.01)	(0.01)
	[0.06]	[0.90]	[0.31]	[0.06]	[0.09]
Pre-treatment Mean	126.5	2.5	0.28	0.06	0.53
Observations	3,881	3,881	3,881	3,881	3,332
Low Ability	31.51	−0.005	−0.09	0.03	0.13
	(5.52)	(0.02)	(0.03)	(0.01)	(0.02)
	[0.24]	[0.90]	[0.36]	[0.29]	[0.20]
Pre-treatment Mean	113.3	2.6	0.32	0.07	0.43
Observations	4,127	4,127	4,127	4,127	3,648

Notes: This table reports heterogeneous treatment effects estimated separately for high-, medium-, and low-ability students. Each coefficient comes from a distinct difference-in-differences regression estimated within an ability group. Standard errors, clustered at the program level, are reported in parentheses. Wild cluster bootstrap p-values are reported in brackets and serve as the basis for statistical inference. Ability terciles are defined using the distribution of high school GPA in the pre-treatment cohorts. ***p < 0.01, **p < 0.05, *p < 0.10, based on wild-bootstrap p-values.

Finally, formal tests for treatment effect heterogeneity fail to reject the null hypothesis that absolute treatment effects are equal across ability groups (Appendix Table A4; bootstrap $p > 0.20$). While this result must be interpreted with caution due to the likely power constraints inherent in the triple-difference specification, particularly given the higher residual variance observed in the low-ability tertile, it is consistent with the substantial positive point estimates observed across all subgroups. Collectively, these findings imply that the policy's benefits were widely distributed across the ability spectrum, rather than being concentrated.

7.7.2 Heterogeneous Treatment Effects: Social Work

The heterogeneous treatment effects for Social Work, summarized in Table 8, suggest that while the reform improved outcomes across the ability distribution, the relative magnitude of these benefits was uneven. In contrast to the progressive convergence observed in Business Administration, the policy's impact in Social Work follows a distinct inverted U-shape with respect to ability, while none of the estimated effects across ability groups achieve statistical significance. While positive point estimates are observed for all groups, the marginal benefit of the deadline was highest for students in the middle of the distribution, who realized substantially larger relative gains than their peers at either tail.

In both the short and long run, credit accumulation increases across all ability groups, but relative gains are largest for medium-ability students. By Semester 4, medium-ability students accumulate nearly 19 percent more credits relative to baseline, compared with roughly 10 percent for both high- and low-ability students. This pattern persists through Semester 7, where medium-ability students again exhibit the largest proportional increase in cumulative credits (15 percent), exceeding the gains observed for high-ability (10 percent) and low-ability (12 percent) students.

Graduation outcomes reinforce this asymmetry. On-time graduation increases most sharply for medium-ability students, with a 124 percent relative increase over a low baseline, compared with 44 percent for high-ability students and 15 percent for low-ability students. Similarly, graduation within nine semesters rises by 27 percent for medium-ability students, while effects are small for high-ability students and slightly negative for low-ability students. Although these estimates are imprecisely estimated, the relative magnitudes consistently point to medium-ability students as the primary beneficiaries of the reform in Social Work.

Table 8: Heterogeneous Treatment Effects by Student Ability: SW

Panel A. Short-Term Effects (Semester 2)

	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
High Ability	8.918	0.023	-0.077*	-0.062
	(2.008)	(0.027)	(0.016)	(0.030)
	[0.114]	[0.499]	[0.076]	[0.278]
Pre-treatment Mean	86.7	2.0	0.18	0.69
Observations	4,080	4,080	4,080	1,106
Medium Ability	14.503	0.011	-0.085	-0.167
	(2.593)	(0.028)	(0.029)	(0.016)
	[0.104]	[0.746]	[0.275]	[0.149]
Pre-treatment Mean	76.5	2.2	0.21	0.66
Observations	3,306	3,306	3,306	1,556
Low Ability	6.578	0.134	-0.062	-0.007
	(2.681)	(0.025)	(0.033)	(0.023)
	[0.520]	[0.348]	[0.487]	[0.760]
Pre-treatment Mean	66.5	2.2	0.26	0.50
Observations	4,050	4,050	4,050	2,838

Panel B. Long-Term Effects (Semester 7)

	Credit Points (5)	GPA (6)	Dropout (7)	On-Time Graduation (8)	Graduated ≤ 9 Semesters (9)
High Ability	15.266	-0.021	-0.065	0.060	0.038
	(3.403)	(0.027)	(0.015)	(0.025)	(0.018)
	[0.115]	[0.536]	[0.105]	[0.227]	[0.304]
Pre-treatment Mean	151.1	1.9	0.19	0.14	0.66
Observations	4,080	4,080	4,080	4,080	3,417
Medium Ability	20.960	-0.066	-0.062	0.090	0.140
	(4.685)	(0.030)	(0.029)	(0.018)	(0.021)
	[0.152]	[0.265]	[0.298]	[0.165]	[0.245]
Pre-treatment Mean	135.9	2.1	0.24	0.07	0.52
Observations	3,306	3,306	3,306	3,306	2,843
Low Ability	13.580	0.087	-0.038	0.011	-0.021
	(4.948)	(0.024)	(0.031)	(0.011)	(0.017)
	[0.492]	[0.374]	[0.536]	[0.523]	[0.414]
Pre-treatment Mean	117.0	2.1	0.31	0.07	0.44
Observations	4,050	4,050	4,050	4,050	3,573

Notes: This table reports heterogeneous treatment effects estimated separately for high-, medium-, and low-ability students. Each coefficient comes from a distinct difference-in-differences regression estimated within an ability group. Standard errors, clustered at the program level, are reported in parentheses. Wild cluster bootstrap p-values are reported in brackets and serve as the basis for statistical inference. Ability terciles are defined using the distribution of high school GPA in the pre-treatment cohorts. ***p < 0.01, **p < 0.05, *p < 0.10, based on wild-bootstrap p-values.

Formal triple-difference tests fail to reject the null hypothesis of equal treatment effects across ability groups (Appendix Table A5), reflecting limited statistical power. Nevertheless, the distribution of relative effects suggests that binding deadlines in Social Work primarily accelerate progress for students in the middle of the ability distribution, without generating adverse effects for lower-ability students.

8. Robustness Checks: Propensity-Score Matched Difference-in-Differences

To assess the sensitivity of the estimates to observable compositional differences, I re-estimate the main specifications using a propensity-score–matched difference-in-differences (matched DiD) approach (Heckman et al., 1997). Although DiD with covariates identifies effects under parallel trends, sizeable demographic gaps—especially the pronounced gender imbalance in Social Work—can inflate residual variance and weaken precision in a small-cluster setting[6]. Matching restricts comparisons to observably similar students, improving overlap and reducing noise.

I estimate propensity scores separately for each program via logit regression of treatment status on high-school GPA, Abitur type, gender, age at matriculation, and cohort fixed effects. Including cohort fixed effects ensures matching occurs within cohorts, preventing cross-period comparisons. I implement radius matching with a caliper of 0.10 (in propensity score standard deviations) and allow replacement, yielding variable matching ratios that optimize covariate balance. Appendix Figures A3 and A4 confirm that matching substantially improves covariate balance for both the BA and SW programs. I then re-estimate Equation (1) in the matched sample using matching weights and the same fixed-effects structure as in the primary specification[7]. Lastly, I conduct Placebo Tests to confirm that the observed effects are not driven by spurious pre-existing trends.

8.1 Matched DiD: Business Administration

For BA, matching confirms the robustness of the primary results and sharpens inference on attrition and graduation outcomes, as reported in Table 9. In the short-term (Panel A), treated

[6] Although DiD with covariates already adjusts for observable differences, linear regression can rely on extrapolation when treated and control groups differ sharply. Matching prevents such extrapolation and improves precision by enforcing overlap, while leaving identification assumptions unchanged.

[7] Unlike the primary analysis, randomization inference is not employed here because it typically treats the matching procedure as fixed, thereby ignoring the variance introduced by estimating propensity scores. The wild cluster bootstrap is preferred in this context as it explicitly accounts for this additional estimation uncertainty and the heteroskedasticity induced by matching weights, while maintaining validity for the small number of clusters.

students accumulated 8.0 additional ECTS credits (bootstrap p=0.001) compared to the estimated 7.2 in the primary DiD specification. GPA remains statistically unchanged (p=0.181), confirming credit gains occurred without quality dilution. While total first-year attrition did not change, the share of eventual dropouts exiting during the first year increased by 9.9 percentage points (bootstrap p=0.007). This significant result, which was imprecise in the primary specification, confirms that the deadline accelerates the exit of poorly matched students without increasing overall dropout rates.

Table 9. Matched DiD Results: Business Administration

Panel A. Short-Term Effects (Semester 2)				
	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
Matched DiD	7.99***	−0.02	−0.02	0.099***
	(0.91)	(0.01)	(0.01)	(0.01)
	[0.000]	[0.18]	[0.24]	[0.007]
Pre-treatment mean	40.27	2.57	0.13	0.52
Effect size (%)	19.86	−0.91	−17.07	18.97
R-squared	0.14	0.15	0.06	0.03
Cohort & Program FE	Yes	Yes	Yes	Yes
Observations	11,842	11,842	11,842	5,754

Panel B. Long-Term Effects (Semester 7)					
	Credit Points (1)	GPA (2)	Dropout (3)	On-Time Graduation (4)	Graduated ≤ 9 Semesters (5)
Matched DiD	24.53***	−0.019	−0.069*	0.049*	0.090**
	(3.18)	(0.009)	(0.01)	(0.01)	(0.01)
	[0.000]	[0.16]	[0.058]	[0.07]	[0.02]
Pre-treatment mean	130.48	2.43	0.26	0.10	0.57
Effect size (%)	18.80	−0.80	−26.80	51.26	15.87
R-squared	0.10	0.15	0.09	0.09	0.06
Cohort & Program FE	Yes	Yes	Yes	Yes	Yes
Observations	11,842	11,842	11,842	11,842	10,178

Notes: Propensity scores are estimated via logit using high school GPA, degree type, gender, age, and cohort. Matching uses radius matching with caliper 0.10, with replacement and common support. Matched DiD estimates are obtained by weighted OLS with cohort and program fixed effects. standard errors are clustered at the program level and reported in parentheses. Wild cluster bootstrap p-values are reported in brackets. ***p < 0.01, **p < 0.05, *p < 0.10, based on wild-bootstrap p-values.

Long-term effects are similarly robust. By Semester 7, treated students earned 24.5 additional credits (bootstrap p = 0.001), aligning closely with the primary estimate. Cumulative dropout falls by 6.9 percentage points (bootstrap p=0.058), moving from statistically insignificant in the main DiD to marginal significance. Finally, the matched specification yields sharper inference for degree completion outcomes: On-time graduation increases by 4.9 percentage points (bootstrap p=0.078), and nine-semester graduation by 9 percentage points (bootstrap p=0.027). These results reinforce the conclusion that the reform accelerated academic progress and improved degree completion without lowering academic standards. The matched event study (Figure 3) reinforces these findings by achieving tighter pre-trend alignment than the baseline model.

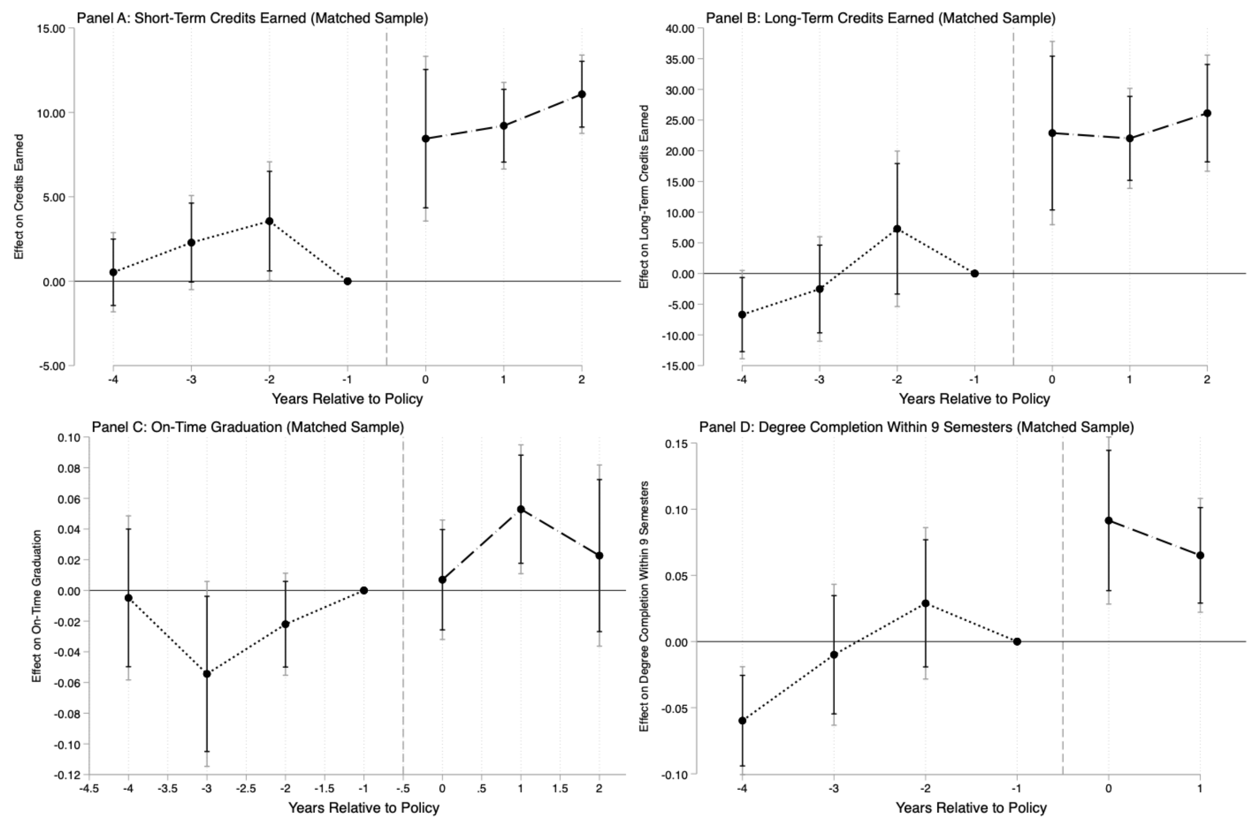

Figure 3: Matched Sample Event Study Graphs: Business Administration *Notes:* This figure plots event-study coefficients for Business Administration using the propensity-score–matched sample, with matching on covariates implemented within cohorts via radius matching. All estimates are relative to the 2015 pre-reform cohort (omitted reference). Point estimates are shown with 95% (grey) and 90% (black) confidence intervals based on standard errors clustered at the program level.

8.2 Matched DiD: Social Work

For Social Work, where treated and control programs differ more sharply in the gender composition, the matched DiD approach delivers substantial precision gains while retaining nearly the entire control group. Table 10 reports the matched DiD estimates. In Semester 4, treated students accumulated 10.1 additional credits (bootstrap p=0.017), a 12.4 percent increase over the pre-treatment mean. This estimate is both larger and more precisely measured than the primary estimate of 7.8 credits (p=0.212), indicating that balancing on observables substantially improves estimation precision[8].

Table 10. Matched DiD Results**:** Social Work

Panel A. Short-Term Effects (Semester 4)

	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
Matched DiD	10.10**	0.08**	−0.06	−0.02
	(2.08)	(0.01)	(0.02)	(0.02)
	[0.016]	[0.015]	[0.12]	[0.49]
Pre-treatment mean	81.28	2.12	0.20	0.84
Effect size (%)	12.43	4.03	−35.51	−2.38
R-squared	0.11	0.33	0.10	0.02
Cohort & Program FE	Yes	Yes	Yes	Yes
Observations	11,432	11,432	11,432	5,500

Panel B. Long-Term Effects (Semester 7)

	Credit Points (1)	GPA (2)	Dropout (3)	On-Time Graduation (4)	Graduated ≤ 9 Semesters (5)
Matched DiD	17.02**	0.02	−0.05	0.057*	0.050
	(3.53)	(0.01)	(0.02)	(0.01)	(0.02)
	[0.015]	[0.22]	[0.14]	[0.07]	[0.22]
Pre-treatment mean	142.64	2.04	0.22	0.11	0.59
Effect size (%)	11.93	1.43	−26.38	52.98	8.51
R-squared	0.11	0.34	0.10	0.10	0.05
Cohort & Program FE	Yes	Yes	Yes	Yes	Yes
Observations	11,432	11,432	11,432	11,432	9,831

Notes: Propensity scores are estimated via logit using high school GPA, degree type, gender, age, and cohort. Matching uses radius matching with caliper 0.10, with replacement and common support. Matched DiD estimates are obtained by weighted OLS with cohort and program fixed effects. standard errors are clustered at the program level and reported in parentheses. Wild cluster bootstrap p-values are reported in brackets. ***p < 0.01, **p < 0.05, *p < 0.10, based on wild-bootstrap p-values.

[8] The matched DiD improves comparability by restricting the sample to students with similar propensity scores, effectively reducing heterogeneity-driven noise. This stabilization of the bootstrap distribution allows previously imprecise effects, especially in the Social Work program, to reach standard significance levels. Ultimately, matching resolves the trade-off between conservative inference and estimation precision by minimizing between-cluster variance.

One cautionary note emerges: matching detects a small, statistically significant decline in short-term GPA (p=0.015), likely reflecting temporary adjustment costs as students accelerated their pace. However, this effect fully dissipates by Semester 7 (p=0.22), indicating no lasting penalty on academic quality.

Long-term outcomes show a similar pattern. By Semester 7, treated students have accumulated 17.0 additional credits (bootstrap $p = 0.015$), and on-time graduation increases by 5.7 percentage points (bootstrap $p = 0.079$), whereas these effects were too imprecise to interpret in the primary DiD. The matched results, therefore, clarify that the earlier statistical uncertainty reflects demographic heterogeneity and limited power, not weak behavioral responses. Overall, matching resolves the ambiguity of the baseline estimates and demonstrates that the delayed deadline generated meaningful and sustained academic momentum.

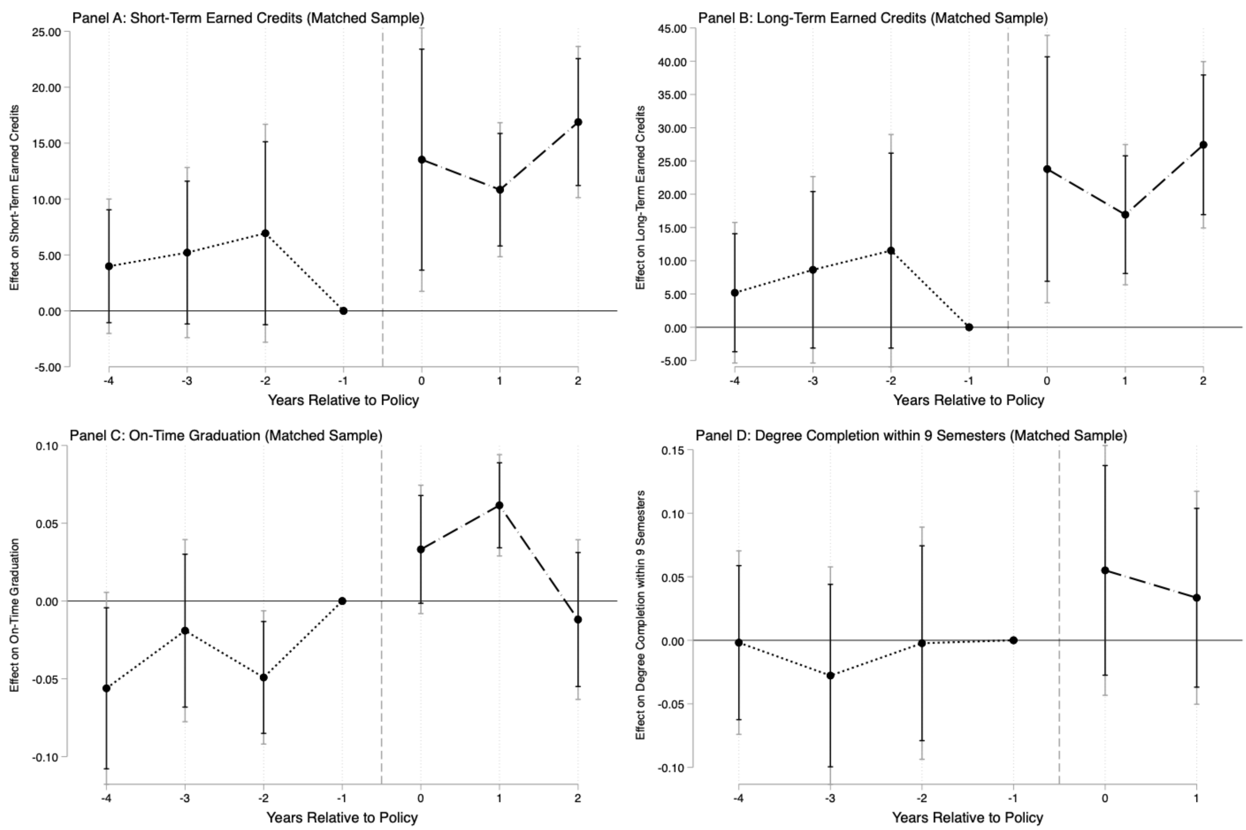

Figure 4: Matched Sample Event Study Graphs: Social Work. *Notes:* This figure plots event-study coefficients for Social Work using the propensity-score–matched sample, with matching on covariates implemented within cohorts via radius matching. All estimates are relative to the 2015 pre-reform cohort (omitted reference). Point estimates are shown with 95% (grey) and 90% (black) confidence intervals based on standard errors clustered at the program level.

Figure 4 presents the dynamic treatment effects for the primary outcomes of interest—credit accumulation and degree completion—using the propensity-score matched sample. By restricting comparisons to observably similar students, the matching procedure significantly stabilizes the pre-treatment trajectories, which now fluctuate more closely around zero than in the primary event study.

8.3 Placebo Difference-in-Differences Test: Business Administration

As an additional robustness check, I implement a placebo difference-in-differences design that assigns the treatment to an earlier, pre-policy period. Specifically, I assume a placebo treatment year of 2014 and restrict the sample to cohorts entering in 2013–2015, treating the 2013 cohort as pre-treatment and the 2014–2015 cohorts as post-treatment. Under the identifying assumption of parallel trends, this specification should yield no systematic effects, as the BA deadline reform had not yet been implemented during this period. Table 11 reports the estimated placebo effects for short-term outcomes measured in semester 2 and long-term outcomes measured in semester 7.

Table 11. Placebo Difference-in-Differences Estimates: Business Administration

Panel A. Short-Term Effects (Semester 2)				
	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
Placebo DID	-0.49	-0.04	-0.02	0
	(0.69)	(0.02)	(0.01)	(0.02)
	[0.86]	[0.63]	[0.71]	[1.00]
Observations	5,065	5,065	5,065	2,465

Panel B. Long-Term Effects					
	Credit Points (1)	GPA (2)	Dropout (3)	On-Time Graduation (4)	Graduated ≤ 9 Semesters (5)
Placebo DID	5.91	-0.06	-0.01	0.02	0.02
	(2.15)	(0.02)	(0.01)	(0.02)	(0.01)
	[0.52]	[0.55]	[0.92]	[0.92]	[0.72]
Observations	5,065	5,065	5,065	5,065	5,065

Notes: The table presents difference-in-differences estimates for a fictitious placebo treatment assigned to the 2014 cohort. The sample is restricted to students from the 2013, 2014, and 2015 cohorts, with the 2013 cohort serving as the pre-treatment baseline. Standard errors clustered at the program level are reported in parentheses. Randomization inference (RI-t) p-values, calculated using 1,000 permutations of the treatment assignment, are reported in square brackets. Significance levels are indicated by *p<0.10$, **p<0.05, *** p<0.01 based on the RI-t p-values.

Across all outcomes, the placebo estimates are small in magnitude and statistically insignificant when inference is conducted using randomization-inference p-values. The absence of detectable placebo effects provides reassurance that the main results are unlikely to be driven by differential pre-trends or spurious cohort-level shocks, lending further support to the causal interpretation of the baseline difference-in-differences estimates.

8.4 Placebo Difference-in-Differences Test: Social Work

Applying the same placebo design to SW, assigning a fictitious treatment year of 2014, yields similarly null results. As shown in Table 12, the estimated placebo effects are statistically insignificant across short-term and long-term outcomes when inference is based on randomization-inference p-values. While some point estimates, particularly for credits earned, are positive in magnitude, they are imprecisely estimated and do not exhibit a systematic pattern across outcomes or time horizons. These findings further support the view that the estimated Social Work effects are unlikely to be driven by pre-existing trends or spurious cohort-level variation.

Table 12. Placebo Difference-in-Differences Estimates: Social Work

Panel A. Short-Term Effects (Semester 4)				
	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
Placebo DID	4.20	0.007	-0.04	-0.003
	(1.2)	(0.02)	(0.01)	(0.02)
	[0.44]	[1.00]	[0.66]	[1.00]
Observations	4,898	4,898	4,898	2,366

Panel B. Long-Term Effects					
	Credit Points (1)	GPA (2)	Dropout (3)	On-Time Graduation (4)	Graduated ≤ 9 Semesters (5)
Placebo DID	6.55	-0.01	-0.04	-0.02	0.07
	(2.28)	(0.02)	(0.01)	(0.02)	(0.01)
	[0.51]	[0.93]	[0.44]	[0.78]	[0.21]
Observations	4,898	4,898	4,898	4,898	4,898

Notes: The table presents difference-in-differences estimates for a fictitious placebo treatment assigned to the 2014 cohort. The sample is restricted to students from the 2013, 2014, and 2015 cohorts, with the 2013 cohort serving as the pre-treatment baseline. Standard errors clustered at the program level are reported in parentheses. Randomization inference (RI-t) p-values, calculated using 1,000 permutations of the treatment assignment, are reported in square brackets. Significance levels are indicated by *p<0.10$, **p<0.05, *** p<0.01 based on the RI-t p-values.

9. Synthetic Difference-in-Differences

To complement the primary analysis and provide robustness to potential violations of parallel trends, I re-estimate treatment effects using Synthetic Difference-in-Differences (SDID), following Arkhangelsky et al. (2021). SDID combines the strengths of synthetic control methods, which construct a weighted average of control units to match the treated unit's pre-treatment trajectory—with difference-in-differences, which accounts for time-invariant unobserved heterogeneity.

The estimator assigns data-driven unit weights to control programs such that their weighted average closely tracks the treated program's pre-treatment outcomes. These weights are determined by minimizing the discrepancy between the treated unit and the synthetic control during the pre-treatment period, subject to regularization that prevents overfitting to idiosyncratic variation. Additionally, SDID assigns time weights that emphasize pre-treatment periods most relevant for extrapolating to the post-treatment period. Crucially, both sets of weights are estimated using only pre-treatment data and held fixed when computing post-treatment outcomes, ensuring that the counterfactual reflects the historical relationship between treated and control units rather than post-treatment realizations. To explicitly account for evolving student composition across cohorts, I incorporate pre-treatment covariates, using the optimized covariate adjustment procedure of Arkhangelsky et al. (2021), which jointly estimates covariate coefficients and unit weights to minimize pre-treatment imbalance. This ensures that the synthetic control reflects the treated program's trajectory net of observable demographic shifts, with inference conducted using a stratified bootstrap procedure to account for uncertainty in the weight estimation.

9.1 Synthetic DiD: Business Administration

Table 13, Panel A, reports the short-term SDID estimates for the BA program. In the short term (Semester 2), treated students accumulated 6.1 additional credits (SE = 1.04, bootstrap $p < 0.001$), a 15% increase relative to the pre-reform mean. This estimate is broadly consistent with the primary DiD (7.2 credits) and matched-DiD (8.0 credits) results, though modestly more conservative, reflecting the optimized reweighting of control units. The reform also altered the timing of exits: early dropout increases by 10.5 percentage points (bootstrap $p < 0.001$), while total dropout remains unchanged, confirming a significant effect on accelerated sorting rather than increased attrition.

Table 13. Synthetic Difference-in-Differences Estimates: Business Administration

Panel A. Short-Term Effects (Semester 2)				
	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
Synthetic DID	6.058***	-0.048*	-0.002	0.105***
	(1.040)	(0.029)	(0.024)	(0.015)
P-value (bootstrap)	0.000	0.098	0.946	0.000
Effect Size (%)	15.04%	-1.87%	-1.24%	20.05%
Pre-Mean (Treated)	40.270	2.572	0.131	0.522
Controls	Yes	Yes	Yes	Yes
Effective Controls	12.73	12.54	12.72	12.88
Student Observations	11,860	11,860	11,860	5,763
Program-Cohort Obs.	98	98	98	98

Panel B. Long-Term Effects (Semester 7)					
	Credit Points (1)	GPA (2)	Dropout (3)	On-Time Graduation (4)	Graduated ≤ 9 Semesters (5)
Synthetic DID	12.525***	0.001	-0.021	0.045***	0.056**
	(3.754)	(0.032)	(0.024)	(0.006)	(0.026)
P-value (bootstrap)	0.001	0.965	0.382	0.000	0.033
Effect Size (%)	9.60%	0.06%	-7.93%	47.15%	9.83%
Pre-Mean (Treated)	130.479	2.429	0.260	0.096	0.571
Controls	Yes	Yes	Yes	Yes	Yes
Effective Controls	12.68	12.53	12.49	12.42	11.27
Student Observations	11,860	11,860	11,860	11,860	10,195
Program-Cohort Obs.	98	98	98	98	84

Notes: This table reports Synthetic Difference-in-Differences estimates following Arkhangelsky et al. (2021). Bootstrap standard errors (resampling control units) are reported in parentheses. All specifications additionally adjust for student-level covariates (high-school GPA, degree type, gender, and age). *Effective Controls* is computed as 1/HHI, where HHI is the Herfindahl index of synthetic control weights. *Effect Size (%)* is computed relative to the treated group's pre-treatment mean. Significance stars based on bootstrap p-values: *** p<0.01, ** p<0.05, * p<0.10.

Long-term effects (panel B) confirm that these early gains persist. Treated students earned 12.5 additional credits (SE = 3.75, bootstrap $p = 0.001$), smaller than the primary DiD estimate (21.9 credits) but precisely estimated. These credit gains translated into faster degree completion: on-time graduation rises by 4.5 percentage points (bootstrap $p < 0.001$), and completion within nine semesters increases by 5.6 percentage points (bootstrap $p = 0.033$). Crucially, both the matched DiD and the synthetic DiD specifications yield robust and statistically significant effects on graduation outcomes, resolving the imprecision observed in the primary framework and providing consistent evidence that the reform accelerated degree completion.

Figure 5 presents the corresponding event-study estimates, which decompose treatment effects by cohort relative to the reform. Pre-treatment coefficients ($t < 0$) fluctuate around zero for all outcomes, with 95 percent confidence intervals spanning the null. This pattern provides visual confirmation that the reweighted synthetic control reduces pre-existing trend differentials. Following the reform ($t \geq 0$), the coefficients shift discretely upward and remain elevated across post-treatment cohorts. For visual reference, Appendix Figure A5 plots the raw outcome trajectories, demonstrating that the synthetic control closely tracks the treated program throughout the pre-reform period before sharply diverging at treatment onset.

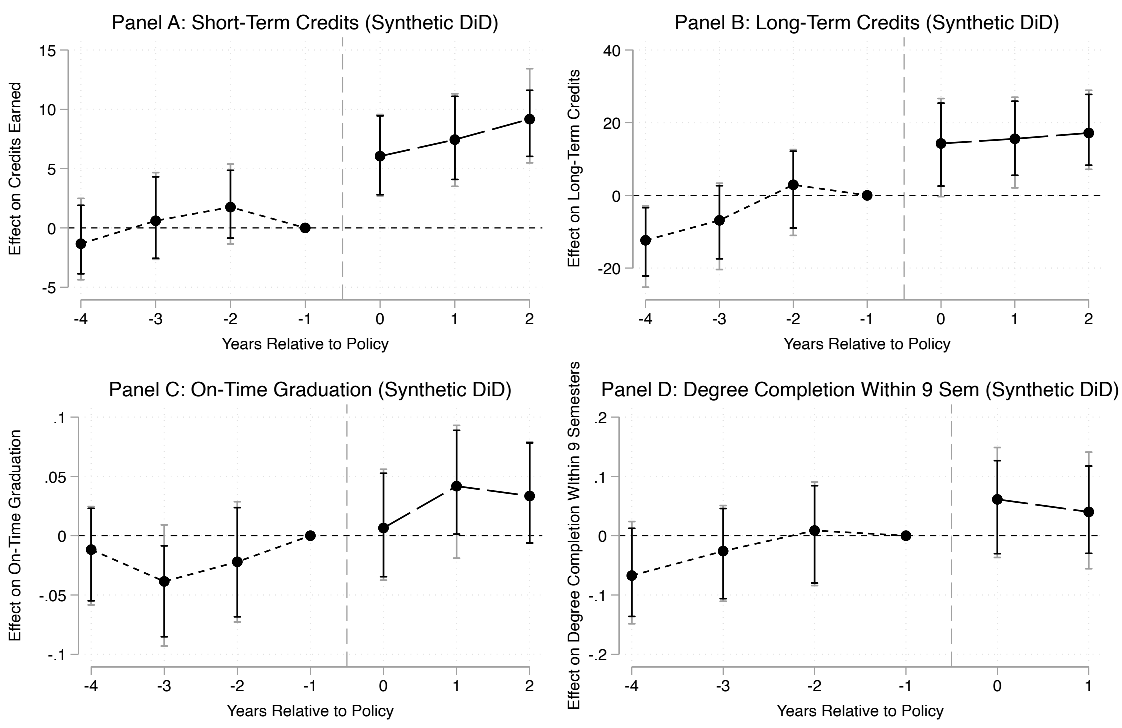

Figure 5: Synthetic Difference-in-Differences Event Study (Business Administration). *Notes*: This figure displays dynamic treatment effects estimated using the Synthetic Difference-in-Differences estimator (Arkhangelsky et al., 2021). Each coefficient represents the normalized difference between the treated Business Administration program and the reweighted synthetic control, with the last pre-treatment cohort (2015, $t = -1$) serving as the reference period. Panels A and B report effects on cumulative credits at Semesters 2 and 7. Panels C and D report effects on on-time graduation and graduation within nine semesters. Vertical bars denote 95 percent confidence intervals derived from a stratified bootstrap procedure that resamples control units.

9.2 Synthetic DiD: Social Work

Table 14 reports the Synthetic Difference-in-Differences estimates for the Social Work program. These results are particularly valuable as they resolve the statistical imprecision observed in the primary unweighted DiD specifications, confirming the robustness of the effects detected in the matched-DiD framework.

Table 14. Synthetic Difference-in-Differences Estimates: Social Work

Panel A. Short-Term Effects (Semester 2)				
	Credit Points (1)	GPA (2)	Dropout (3)	Early Dropout (4)
Synthetic DID	8.648***	0.044	-0.044**	-0.068*
	(2.398)	(0.028)	(0.018)	(0.037)
P-value (bootstrap)	0.000	0.122	0.018	0.070
Effect Size (%)	10.64%	2.07%	-22.13%	-10.57%
Pre-Mean (Treated)	81.279	2.116	0.197	0.64
Controls	Yes	Yes	Yes	Yes
Effective Controls	12.68	12.17	12.82	12.31
Student Observations	11,437	11,437	11,437	5,500
Program-Cohort Obs.	98	98	98	98

Panel B. Long-Term Effects (Semester 7)					
	Credit Points (1)	GPA (2)	Dropout (3)	On-Time Graduation (4)	Graduated ≤ 9 Semesters (5)
Synthetic DID	11.454***	0.005	-0.011	0.035***	0.071***
	(3.704)	(0.023)	(0.022)	(0.012)	(0.028)
P-value (bootstrap)	0.002	0.810	0.619	0.003	0.010
Effect Size (%)	8.03%	0.27%	-5.03%	32.46%	12.02%
Pre-Mean (Treated)	142.643	2.038	0.218	0.108	0.593
Controls	Yes	Yes	Yes	Yes	Yes
Effective Controls	12.84	12.56	12.90	12.44	12.28
Student Observations	11,437	11,437	11,437	11,437	9,834
Program-Cohort Obs.	98	98	98	98	84

Notes: This table reports Synthetic Difference-in-Differences estimates following Arkhangelsky et al. (2021). Bootstrap standard errors (resampling control units) are reported in parentheses. All specifications additionally adjust for student-level covariates (high-school GPA, degree type, gender, and age). Effective Controls is computed as 1/HHI, where HHI is the Herfindahl index of synthetic control weights. Effect Size (%) is computed relative to the treated group's pre-treatment mean. Significance stars based on bootstrap p-values: *** p<0.01, ** p<0.05, * p<0.10.

In the short run (Panel A), treated students accumulated 8.6 additional credits (SE = 2.40, $p < 0.001$), a 10.6% increase relative to the synthetic counterfactual. Panel B confirms that this initial momentum translated into sustained long-term gains. By Semester 7, treated students earned 11.5 additional credits ($p = 0.002$). Most crucially, the SDID weighting strategy resolves the noise that obscured degree completion effects in the unweighted specifications. On-time graduation increased significantly by 3.5 percentage points ($p = 0.003$), and completion within nine semesters rose by 7.1 percentage points ($p = 0.010$). Notably, there were no detectable long-term effects on GPA or overall dropout rates, indicating that the policy successfully increased academic pace without compromising performance standards or inducing late-stage attrition.

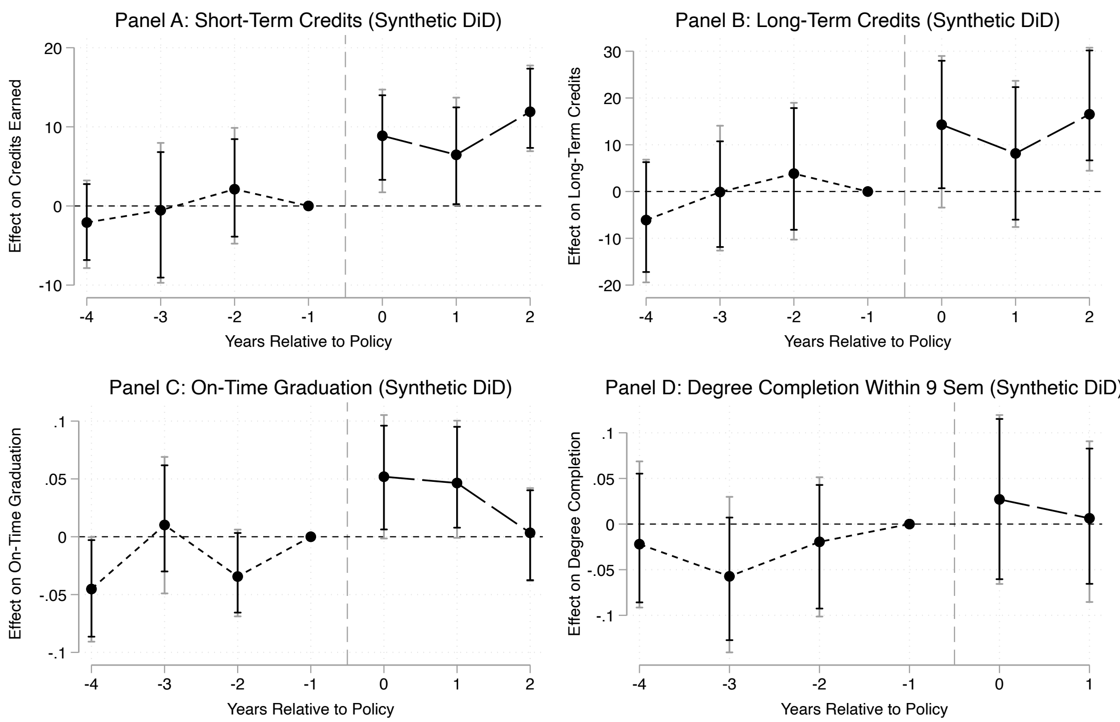

Figure 6: Synthetic Difference-in-Differences Event Study (Social Work). *Notes*: This figure displays dynamic treatment effects estimated using the Synthetic Difference-in-Differences estimator (Arkhangelsky et al., 2021). Each coefficient represents the normalized difference between the treated Social Work program and the reweighted synthetic control, with the last pre-treatment cohort (2015, $t = -1$) serving as the reference period. Panels A and B report effects on cumulative credits at Semesters 2 and 7. Panels C and D report effects on on-time graduation and graduation within nine semesters. Vertical bars denote 95 percent confidence intervals derived from a stratified bootstrap procedure that resamples control units.

Figure 6 presents the dynamic treatment effects. Although the raw pre-treatment trajectories display some stochastic variation—unsurprising given the program-level aggregation—the event-study estimates indicate that the reweighted synthetic control effectively absorbs these fluctuations. Pre-treatment coefficients are small and statistically indistinguishable from zero for all $t<0$, confirming that the estimator achieves a close pre-treatment match and constructs a credible counterfactual despite underlying noise in the data. Appendix Figure A6 displays the corresponding unweighted trends.

10. Discussion and Conclusion

This study demonstrates that mandatory course-completion deadlines substantially improve student outcomes in Germany's tuition-free, high-autonomy higher education system. Exploiting policy reforms where two programs introduced binding deadlines in 2016, I find deadlines increased early credit accumulation by 10–18 percent, long-term accumulation by 9–17 percent, reduced cumulative dropout by roughly 6 percentage points, and raised timely graduation by 5–9 percentage points. These gains emerged without lowering GPA and without increasing overall attrition, indicating a genuine expansion of effort rather than quality dilution or negative selection. Both an early, targeted, high-stakes design (BA) and a later, comprehensive, moderate-stakes design (SW) generated meaningful improvements, though the former produced sharper and more precisely estimated effects. The convergence of results across unweighted, matched, and synthetic specifications confirms that the observed improvements reflect a systematic behavioral response rather than idiosyncratic trend drift.

Mechanisms: Commitment and Momentum—The results align closely with behavioral models of present bias and procrastination (O'Donoghue and Rabin, 1999, 2001). In environments with weak short-run consequences for delay and high autonomy, students have incentives to postpone costly effort. This dynamic is pervasive: approximately 80% to 95% of college students procrastinate, with nearly half doing so chronically in ways that impair performance (Steel, 2007). Binding deadlines raise the immediate cost of inaction and therefore function as externally imposed commitment devices, analogous to penalty-based contracts shown to increase worker effort in labor markets (Ariely and Wertenbroch 2002; Kaur, Kremer, and Mullainathan 2015). Given the widespread prevalence of procrastination, these interventions hold promise for benefiting a broad student population, not just those with acute self-regulation challenges.

Consistent with this mechanism, recent evidence from a field experiment demonstrates that offering students commitment devices directly increases the number of credits they earn (Himmler, Jäckle, and Weinschenk 2019). Similarly, Patterson (2018) demonstrates that external pacing tools significantly improve persistence in unstructured learning environments by preventing students from falling behind. In the present setting, stable GPAs and unchanged overall dropout rates reinforce this interpretation: students responded to deadlines not by reallocating effort or strategically exiting, but by expanding total effort in a way consistent with commitment-driven behavioral change.

The effort interpretation also finds support in Ostermaier's (2018) analysis of the same institutional type: when a German university removed a binding deadline for introductory modules, students increasingly submitted blank exams and strategically postponed effort, thereby extending their study duration. The reforms studied here reverse this dynamic. Furthermore, Müller (2023) uses a regression discontinuity design to show that students who narrowly fail an early exam are significantly more likely to graduate than those who narrowly pass. This indicates that early failure signals, when recovery remains feasible, act as corrective 'wake-up call' rather than sources of discouragement. The deadline reforms likely operate through a similar channel: by establishing clear, early performance thresholds, they make underperformance salient and redirect effort toward degree completion instead of inducing exit.

Crucially, the early acceleration triggered self-reinforcing academic momentum. Short-run credit gains persisted proportionally into later semesters, translating into higher completion rates. Three complementary channels likely operate: (i) accumulation of academic human capital and study habits that compound over time (Stinebrickner and Stinebrickner, 2008, 2014); (ii) strengthened social and institutional integration through regular engagement (Tinto, 2012; Astin, 1999); and (iii) psychological commitment via sunk costs and enhanced self-efficacy (Arkes and Blumer, 1985; Bandura, 1997). The persistence of proportional effects rules out simple front-loading and provides some of the first quasi-experimental evidence that institutions can actively generate momentum rather than merely observe its correlates (Attewell et al., 2012; Belfield et al., 2016).

Policy Design and Institutional Context—These findings directly relate to the debate on academic dismissal policies. Some evidence from the Dutch system shows that strict credit thresholds mainly function through selection, causing dropout to occur earlier without improving graduation rates (Arnold 2015; Cornelisz et al. 2020). In contrast, the results reveal clear incentive effects: credits and graduation rates increase, GPA stays stable, and overall dropout does not rise. Such patterns are inconsistent with cream-skimming and align with genuine behavioral responses. The only selective effect is that students who would have dropped out anyway leave earlier, indicating better information rather than the exclusion of potentially successful students. This distinction has important welfare implications. Recent evidence suggests that students near dismissal thresholds who persist to graduation earn substantial returns (Ost, Pan, and Webber, 2018), indicating that policies facilitating completion rather than exit generate meaningful human capital gains even for marginal students.

The contrast with the Dutch results likely reflects differences in policy design. The German deadlines function as a targeted and fully recoverable form of academic dismissal: they impose meaningful progression requirements that strengthen student effort, yet remain mild enough to avoid the discouragement and attrition effects often seen under harsher, system-wide dismissal regimes (Lindo et al., 2010).

To understand which specific design features drive these incentives, it is necessary to examine the interaction of timing, scope, and flexibility. A comparison of the BA and SW reforms reveals distinct approaches: BA imposed early, targeted deadlines with tight constraints, while SW employed broader and later requirements with more leniency in implementation. Interestingly, these features appear to counterbalance one another; the breadth of the later deadline was diluted by its flexibility, whereas the narrow scope of the early deadline was amplified by its immediacy. The fact that both reforms produced similar improvements suggests that, within highly flexible systems, early intervention can effectively substitute for comprehensiveness by shaping behavior during the critical period of study habit formation.

Equity and Economic Magnitude—A frequent concern is that performance standards disproportionately harm less-prepared students (Scott-Clayton, 2015). The heterogeneity analysis provides reassuring evidence to the contrary: treatment effects were large and positive across the entire high-school-GPA distribution, with effect size often largest (though noisier) for low-ability students. Deadlines appear to function as a universal pacing device that reduces behavioral frictions for all students. The findings are consistent with experimental findings from CUNY's ASAP replication in Ohio, where structured support improved outcomes equally for students with and without developmental needs (Scrivener et al., 2015).

A central advantage of the deadline reforms is that they generate substantial academic gains at virtually zero marginal cost. Unlike resource-intensive interventions such as personalized coaching (Bettinger and Baker 2014), summer bridge programs (Barnett et al. 2012), or large-scale financial aid expansions (Goldrick-Rab et al. 2016), adjusting progression rules requires only administrative changes. In a publicly funded system, such structural nudges can yield unusually high returns. As shown by Weiss (2019), who emphasizes that low-cost institutional structures, such as clear milestones and timely progress requirements, can improve outcomes for a wide range of students.

A back-of-the-envelope calculation illustrates the magnitude. Consider the Business Administration program, where the reform increased graduation within nine semesters by 7 percentage points. Public expenditure on tertiary education in Germany averages roughly €11,700 per student-year (OECD 2023), and the median starting salary for business graduates is about €45,000 (Heming et al., 2020). If the marginal graduates enter the labor market just one semester earlier, the combined public savings and additional earnings amount to approximately €28,000 per student. For a typical cohort of 250 students, this implies total social benefits of around €500,000 per cohort, generated entirely by a low-cost structural change that reduces excess time-to-degree.

Limitations and Future Research—Several limitations qualify the interpretation of these findings and point toward productive avenues for future work. First, the analysis relies on reforms within a single university. While the institutional setting is representative of Germany's flexible, low-tuition system, the single-treated-unit design inherently limits external validity. Second, the comparison group consists predominantly of STEM programs. Although event-study tests support the parallel trends assumption, I cannot rule out unmeasured, time-varying shocks specific to STEM fields, such as labor market shifts or funding changes. Future research should exploit variation in deadline policies within the same academic field across multiple institutions to further isolate the treatment effect from sectoral trends.

Third, identification relies on administrative outcomes and therefore cannot disentangle the underlying behavioral mechanisms. The patterns are consistent with momentum and commitment-device channels, but direct evidence on study time or effort is unavailable. Furthermore, the net impact on well-being is ambiguous: deadlines may increase short-term pressure while simultaneously relieving the chronic stress of procrastination. Future research should combine administrative records with survey data, digital traces, or well-being metrics to definitively isolate these pathways and evaluate the full welfare trade-offs. Finally, linking education records to social security earnings data is essential to determine whether the reform affects long-run labor market success (e.g., through habit formation) or simply shifts the timing of entry without altering wage trajectories.

The COVID-19 pandemic represents a shock for the later cohorts in the sample. Campus closures and the transition to remote instruction began in March 2020, coinciding with the fourth semester of the 2018 cohort and the sixth semester of the 2017 cohort. The difference-in-differences framework accounts for such shocks under the assumption that pandemic-related disruptions affected treated and control programs symmetrically within cohorts. However, because STEM-intensive control programs rely more heavily on laboratory-based instruction that was differentially altered during the pandemic, the long-run estimates could be interpreted as conservative lower bounds if disruptions were not symmetric across programs. Specifically, if the cancellation or substitution of laboratory components during lockdowns effectively lowered the threshold for earning credits in control fields relative to the treated programs, the comparison group's outcomes would be mechanically inflated. Such a differential impact would attenuate the estimated treatment effects, implying that the reported results represent a conservative lower bound of the true impact of progression deadlines. Crucially, the university suspended the strict enforcement of progression deadlines during the pandemic semesters, which likely weakened the policy’s incentive effect for the affected cohorts.

In conclusion, this study demonstrates that simple structural policies can substantially improve outcomes in flexible academic settings where weak short-run incentives often lead to drift. By replacing open-ended timelines with clear, recoverable milestones, universities provide necessary commitment devices that activate early momentum. Crucially, these gains do not compromise equity or rigor: deadlines improved progress across the entire ability distribution without increasing attrition. More broadly, the results highlight the central role of institutional design in shaping educational achievement. As higher education faces growing demands for efficiency, these findings suggest that calibrated deadlines offer a scalable, zero-cost tool to support student success without sacrificing access.

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Appendix A: Further Tables and Figures

Table A1. Event-Study Difference-in-Differences Estimates: Business Administration

	Short-term Effects			Long-term Effects					
	Credits Sem 2 (1)	GPA Sem 2 (2)	Dropout Sem 2 (3)	Early Dropout (4)	Credits Sem 7 (5)	GPA Sem 7 (6)	Dropout Sem 7 (7)	On-Time Graduation (8)	Graduated ≤ 9 Semesters (9)
t-4 Coefficient	-0.2818	-0.0223	0.0465	-0.1098**	-9.6424**	-0.0052	0.0436*	0.0042	-0.0745***
	(1.2708)	(0.0172)	(0.0278)	(0.0449)	(4.0040)	(0.0131)	(0.0229)	(0.0222)	(0.0176)
t-3 Coefficient	1.1493*	-0.0207	0.0203**	-0.1336***	-6.4534***	0.020.	0.0120	-0.0315	-0.0265*
	(0.5536)	(0.0217)	(0.0087)	(0.0217)	(1.7826)	(0.0229)	(0.0119)	(0.0249)	(0.0139)
t-2 Coefficient	1.2801	-0.1063***	-0.0017	-0.1253***	-0.6416.	-0.0695***	0.0063.	-0.0264	-0.0112
	(0.7889)	(0.0154)	(0.0233)	(0.0239)	(3.6340)	(0.0159)	(0.0199)	(0.0159)	(0.0142)
t0 Coefficient	5.4828***	-0.0860**	0.0407	-0.0551**	12.7664***	-0.0420	-0.0261.	-0.0004	0.0423***
	(1.3174)	(0.0345)	(0.0262)	(0.0222)	(3.8458)	(0.0284)	(0.0235)	(0.0177)	(0.0143)
t+1 Coefficient	7.3980***	-0.0642*	0.0039	-0.1054***	16.1375***	-0.0296	-0.0185.	0.0470**	0.0352**
	(1.0533)	(0.0320)	(0.0242)	(0.0255)	(3.8405)	(0.0257)	(0.0196)	(0.0206)	(0.0159)
t+2 Coefficient	9.9875***	-0.0194	-0.0212	0.0168.	22.4770***	-0.0093	-0.0677**	0.0305	N/A
	(1.0706)	(0.0283)	(0.0239)	(0.0283)	(4.2744)	(0.0231)	(0.0232)	(0.0311)	N/A
Pre-treatment Mean	40.2702	2.5724	0.1305	0.5217	130.4793	2.4285	0.2597	0.0957	0. 5705
R-squared	0.3084	0.3175	0.0940	0.0344	0.2449	0.3474	0.1961	0.1537	0.2146
Observations	11,877	11,877	11,877	5,779	11,877	11,877	11,877	11,877	10,211
Panel B: Parallel Trends Test									
Joint Test P-val.	0.5185	0.0641	0.4665	0.2643	0.3183	0.0961	0.5556	0.4114	0.3232

Notes: This table reports event study estimates using the full trend group. The policy was implemented for the 2016 cohort; the 2015 cohort is used as the reference group, so all coefficients are relative to this baseline. "Graduated within 9 Semesters" uses cohorts from 2012 to 2017; the 2018 cohort is excluded because students could only be observed through their 8th semester. t–2 corresponds to two cohorts prior to treatment, t0 is the treatment cohort, t+1 is the 2017 cohort, etc.. Panel A reports the estimated coefficients with standard errors in parentheses. Panel B provides p-values for joint tests of pre-treatment coefficients (t–4 through t–1), which assess the parallel trends assumption; "Pass" indicates that the null hypothesis of equal pre-trends cannot be rejected at conventional levels. This assumption is formally tested using a joint hypothesis test of pre-treatment coefficients. Parallel Trends Test Specification: H_0: β_{t-4} = β_{t-3} = β_{t-2} = 0. Parallel trends testing employs a wild cluster bootstrap with Webb weights (999 replications) to test the joint null hypothesis. The bootstrap method is essential given the small number of study program clusters, as standard asymptotic tests can provide unreliable p-values when cluster counts are low. For most outcomes, the parallel trends assumption holds, while GPA outcomes show marginal evidence of differential pre-trends.

Table A2. Event-Study Difference-in-Differences Estimates: SW

	Short-term Effects			Long-term Effects					
	Credits Sem 2 (1)	GPA Sem 2 (2)	Dropout Sem 2 (3)	Early Dropout (4)	Credits Sem 7 (5)	GPA Sem 7 (6)	Dropout Sem 7 (7)	On-Time Graduation (8)	Graduated ≤ 9 Semesters (9)
t-4 Coefficient	2.6135.	-0.1137***	0.0072.	-0.0331.	1.6716	-0.0725***	0.0101	-0.0320	0.1020
	(2.1387)	(0.0142)	(0.0256)	(0.0254)	(3.9746)	(0.0128)	(0.0229)	(0.0221)	(0.1042)
t-3 Coefficient	-3.2954***	-0.0568**	0.0283.	-0.0374.	-5.1197**	-0.0228	0.0270*	0.0083	0.2137**
	(1.0399)	(0.0208)	(0.0176)	(0.0225)	(2.1612)	(0.0215)	(0.0141)	(0.0253)	(0.0718)
t-2 Coefficient	1.8837.	-0.1037***	-0.0232.	-0.1000***	3.0926	-0.0748***	-0.0332	-0.0408**	0.2848***
	(1.8345)	(0.0156)	(0.0264)	(0.0236)	(3.6299)	(0.0160)	(0.0199)	(0.0159)	(0.0807)
t0 Coefficient	5.8738**	0.0323.	-0.0399.	-0.1403***	10.5449**	0.0102	-0.0151	0.0381*	-0.1696*
	(2.3556)	(0.0279)	(0.0247)	(0.0166)	(3.9035)	(0.0297)	(0.0242)	(0.0182)	(0.0895)
t+1 Coefficient	5.8112**	-0.0146.	-0.0153.	-0.0391.	7.0606*	-0.0288	-0.0037	0.0502**	-0.2849**
	(2.1888)	(0.0265)	(0.0237)	(0.0224)	(3.7596)	(0.0267)	(0.0196)	(0.0202)	(0.1050)
t+2 Coefficient	12.5624***	-0.0285.	-0.0794***	-0.0229.	19.0120***	-0.0672***	-0.0667**	0.0045	N/A
	(2.5656)	(0.0226)	(0.0260)	(0.0224)	(4.2860)	(0.0225)	(0.0233)	(0.0311)	N/A
Pre-treatment Mean	81.2789	2.1161	0.1967	0.8400	142.6429	2.0376	0.2177	0.1077	8.4023
R-squared	0.2730	0.4162	0.1762	0.0363	0.2502	0.4392	0.2023	0.1544	0.1892
Observations	11454	11454	11454	5516	11454	11454	11454	11454	4718
Panel B: Parallel Trends Test									
Joint Test P-val.	0.4064	0.1532	0.5005	0.4384	0.3213	0.4685	0.2893	0.3213	0.3303

Notes: This table reports event study estimates using the full trend group. The policy was implemented for the 2016 cohort; the 2015 cohort is used as the reference group, so all coefficients are relative to this baseline. "Graduated within 9 Semesters" uses cohorts from 2012 to 2017; the 2018 cohort is excluded because students could only be observed through their 8th semester. t–2 corresponds to two cohorts prior to treatment, t0 is the treatment cohort, t+1 is the 2017 cohort, etc.. Panel A reports the estimated coefficients with standard errors in parentheses. Panel B provides p-values for joint tests of pre-treatment coefficients (t–4 through t–1), which assess the parallel trends assumption; "Pass" indicates that the null hypothesis of equal pre-trends cannot be rejected at conventional levels. This assumption is formally tested using a joint hypothesis test of pre-treatment coefficients. Parallel Trends Test Specification: H_0: β_{t-4} = β_{t-3} = β_{t-2} = 0. Parallel trends testing employs a wild cluster bootstrap with Webb weights (999 replications) to test the joint null hypothesis. The bootstrap method is essential given the small number of study program clusters, as standard asymptotic tests can provide unreliable p-values when cluster counts are low. For most outcomes, the parallel trends assumption holds, while GPA outcomes show marginal evidence of differential pre-trends.

Table A3: Cross-Program Comparison of Treatment Effects

	Credits (1)	GPA (2)	Dropout (3)
BA: Semester 2 (Primary Binding)	7.154***	-0.018	-0.011
	(0.833)	(0.023)	(0.017)
	[0.001]	[0.934]	[0.924]
BA: Semester 4 (For Comparison)	12.04	-0.022	-0.047
	(1.799)	(0.018)	(0.021)
	[0.124]	[0.868]	[0.469]
SW: Semester 2 (For Comparison)	3.887	0.115*	-0.052
	(0.877)	(0.023)	(0.017)
	[0.266]	[0.080]	[0.324]
SW: Semester 4 (Primary Binding)	7.796	0.062	-0.049
	(1.868)	(0.018)	(0.021)
	[0.266]	[0.158]	[0.469]
Controls	Yes	Yes	Yes
Cohort & Program FE	Yes	Yes	Yes
Observations	11,454	11,454	11,454

Notes: This table compares treatment effects at equivalent time horizons (2 semesters) but different positions relative to the deadline. Standard errors clustered at the study-program level (15 clusters) are in parentheses. Randomization inference (RI-t) p-values are reported in square brackets. Significance stars are based on RI-t: *** p<0.01, ** p<0.05, * p<0.10. The Business Administration (BA) deadline binds at Semester 2. The Social Work (SW) deadline binds in Semester 4.

Table A4. Heterogeneous Treatment Effects: Triple-Difference Estimates (BA)

Outcome	High Ability (1)	Medium Ability (2)	Low Ability (3)	Joint HTE P-val (4)	Diff. Pre-trend H – M (5)	Diff. Pre-trend L – M (6)
Panel A. Short-Term Effects (Semester 2)						
Credit Points	9.003**	6.372**	10.815	0.267	0.286	0.364
	(1.038)	(0.826)	(1.562)	N=11,877		
	[0.010]	[0.032]	[0.265]			
GPA	0.057	0.027	0.002	0.562	0.665	0.632
	(0.039)	(0.026)	(0.025)	N=11,877		
	[0.390]	[0.462]	[0.947]			
Dropout	-0.055	-0.005	-0.024	0.245	0.143	0.352
	(0.017)	(0.015)	(0.031)	N=11,877		
	[0.189]	[0.729]	[0.553]			
Early Dropout	0.114	0.054	0.088	0.353	0.218	0.367
	(0.033)	(0.016)	(0.02)	N=5,779		
	[0.234]	[0.274]	[0.396]			
Panel B. Long-Term Effects (Semester 7)						
Credit Points	31.083***	20.148*	29.32	0.262	0.457	0.48
	(3.216)	(3.727)	(5.815)	N=11,877		
	[0.007]	[0.067]	[0.302]			
GPA	-0.077	-0.001	-0.009	0.398	0.46	0.469
	(0.025)	(0.025)	(0.025)	N=11,877		
	[0.284]	[0.959]	[0.755]			
Dropout	-0.117**	-0.043	-0.082	0.231	0.271	0.427
	(0.015)	(0.024)	(0.035)	N=11,877		
	[0.011]	[0.306]	[0.416]			
On-Time Graduation	0.049	0.071**	0.029	0.38	0.14	0.144
	(0.026)	(0.012)	(0.011)	N=11,877		
	[0.294]	[0.047]	[0.380]			
Graduated ≤ 9 Semesters	0.109*	0.083*	0.117	0.531	0.293	0.459
	(0.017)	(0.017)	(0.023)	N=10,211		
	[0.053]	[0.096]	[0.287]			

Notes: Columns 1–3 report the treatment effects for each ability group estimated using separate DiD regressions in Business Administration. Standard errors clustered at the program level are reported in parentheses. Wild cluster bootstrap p-values are reported in brackets. "Joint HTE" reports the bootstrap p-value for the null hypothesis that treatment effects are equal across all three ability groups. "Diff. Pre-trend" reports the p-value for a joint test of parallel pre-treatment trends in ability gaps relative to the medium-ability group. Early Dropout is conditional on eventual dropout. Graduated ≤ 9 Semesters excludes the 2018 cohort. Significance stars based on bootstrap p-values: *** p<0.01, ** p<0.05, * p<0.10.

Table A5. Heterogeneous Treatment Effects: Triple-Difference Estimates (SW)

Panel A. Short-Term Effects (Semester 2)						
	High Ability (1)	Medium Ability (2)	Low Ability (3)	Joint HTE P-val (4)	Diff. Pre-trend H – M (5)	Diff. Pre-trend L – M (6)
Credit Points	10.318*	14.095	7.997	0.367	0.521	0.401
	(2.003)	(2.704)	(3.156)	N: 11,454		
	[0.033]	[0.132]	[0.576]			
GPA	0.011	-0.000	0.146	0.798	0.096	0.649
	(0.027)	(0.027)	(0.029)	N: 11,454		
	[0.741]	[0.995]	[0.387]			
Dropout	-0.088*	-0.082	-0.075	0.862	0.496	0.479
	(0.016)	(0.028)	(0.034)	N: 11,454		
	[0.020]	[0.271]	[0.524]			
Early Dropout	-0.072	-0.154	-0.005	0.483	0.133	0.448
	(0.032)	(0.018)	(0.024)	N: 5,516		
	[0.257]	[0.356]	[0.849]			
Panel B. Long-Term Effects (Semester 7)						
Credit Points	17.542*	20.479	16.173	0.574	0.556	0.427
	(3.277)	(4.818)	(5.540)	N: 11,454		
	[0.038]	[0.155]	[0.562]			
GPA	-0.034	-0.073	0.096	0.440	0.165	0.725
	(0.026)	(0.029)	(0.028)	N: 11,454		
	[0.377]	[0.276]	[0.387]			
Dropout	-0.077*	-0.062	-0.059	0.660	0.563	0.413
	(0.015)	(0.029)	(0.032)	N: 11,454		
	[0.035]	[0.282]	[0.562]			
On-Time Graduation	0.067	0.092	0.001	0.403	0.568	0.486
	(0.026)	(0.015)	(0.012)	N: 11,454		
	[0.201]	[0.209]	[0.953]			
Graduated ≤ 9 Semesters	0.049	0.121	-0.013	0.247	0.551	0.197
	(0.017)	(0.022)	(0.020)	N: 9,851		
	[0.184]	[0.233]	[0.567]			

Notes: Columns 1–3 report the treatment effects for each ability group estimated using separate DiD regressions. Standard errors clustered at the program level are reported in parentheses. Wild cluster bootstrap p-values are reported in brackets. "Joint HTE" reports the bootstrap p-value for the null hypothesis that treatment effects are equal across all three ability groups. "Diff. Pre-trend" reports the p-value for a joint test of parallel pre-treatment trends in ability gaps relative to the medium-ability group; values above 0.10 indicate no evidence of differential pre-trends. Early Dropout is conditional on eventual dropout (N=5,779). Graduated ≤ 9 Semesters excludes the 2018 cohort. Significance stars based on bootstrap p-values: *** p<0.01, ** p<0.05, * p<0.10.

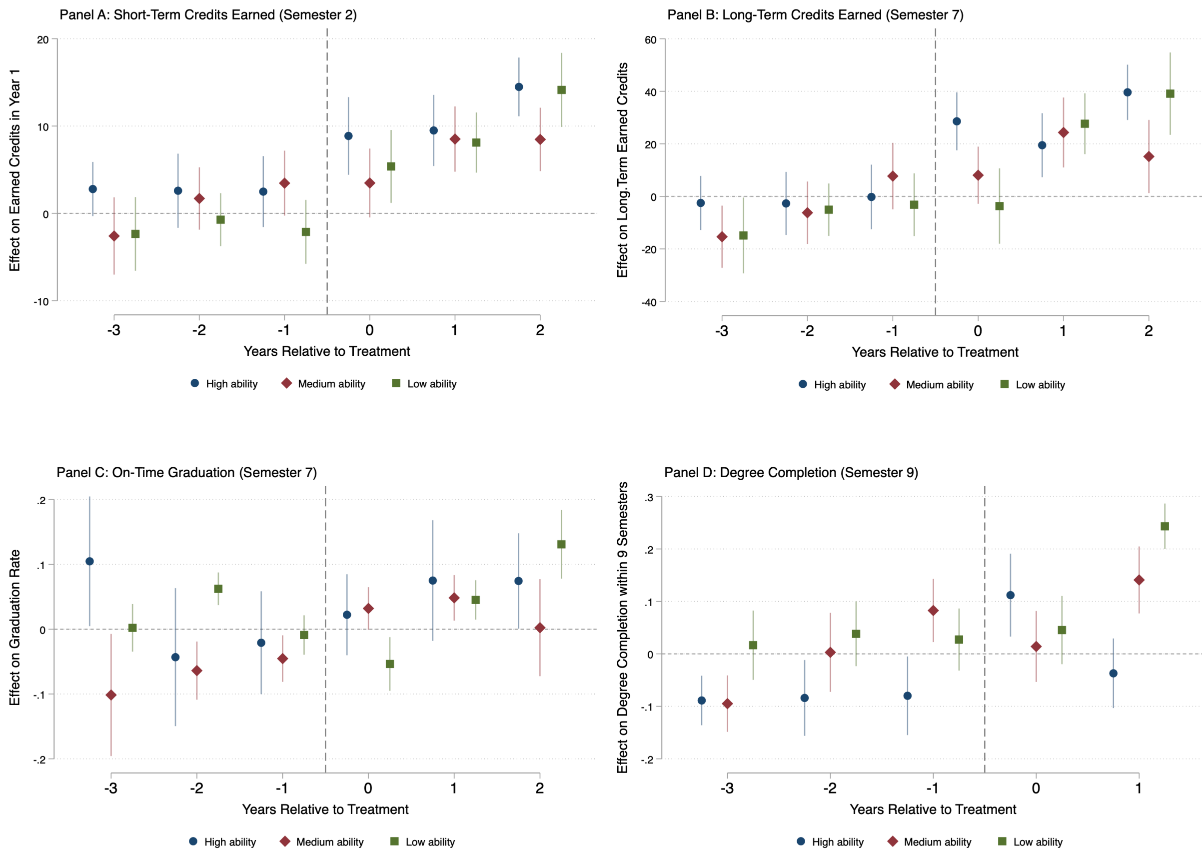

Figure A1. Event-Study Estimates by Ability Group: Business Administration

Notes: This figure reports event-study estimates of the effects of mandatory course-completion deadlines by student ability group. Panels display outcomes for short-term credits earned (Semester 2), long-term credits earned (Semester 7), on-time graduation (Semester 7), and graduation within nine semesters. Ability groups are defined by terciles of high school GPA in the pre-treatment cohorts. Points represent estimated coefficients relative to the 2015 cohort (t = 0), with vertical bars indicating 95 percent confidence intervals based on wild cluster bootstrap standard errors clustered at the program level. Negative values indicate outcomes below the reference cohort mean.

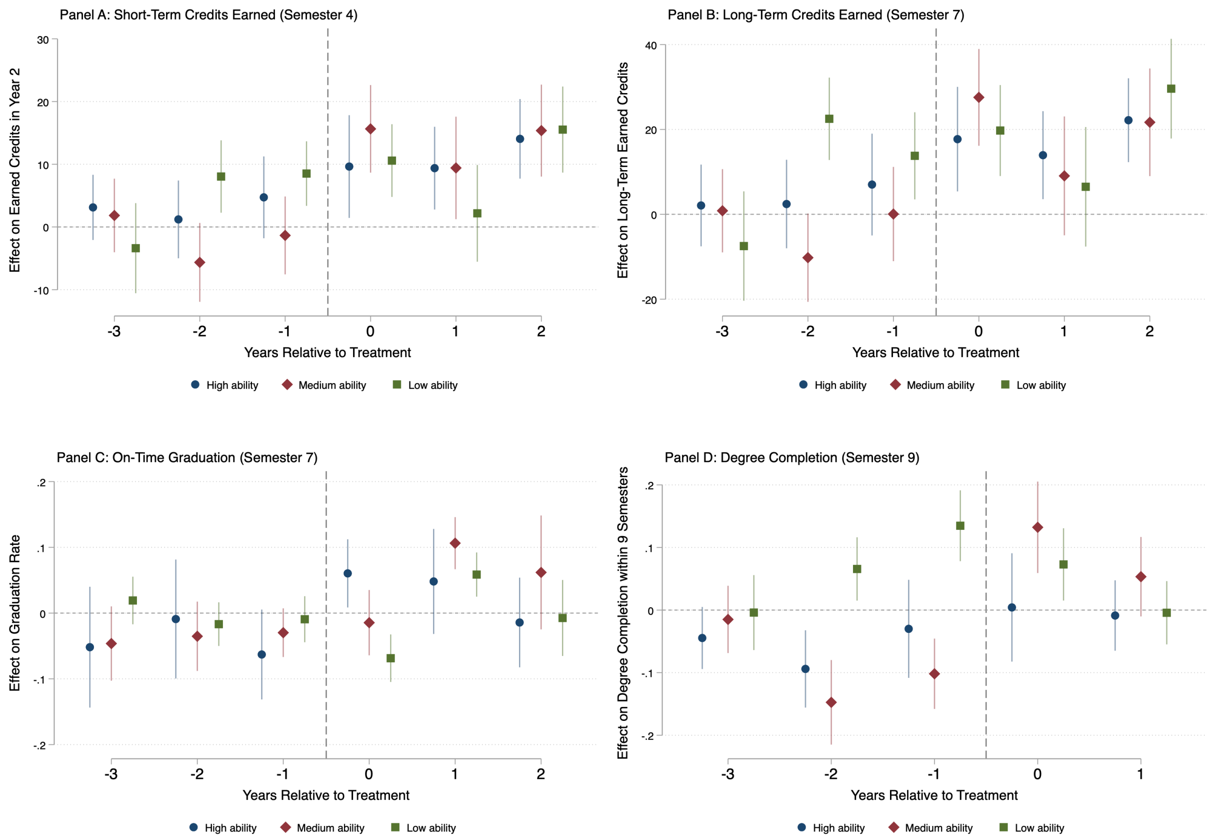

Figure A2. Event-Study Estimates by Ability Group: Social Work

Notes: This figure reports event-study estimates of the effects of mandatory course-completion deadlines by student ability group. Panels display outcomes for short-term credits earned (Semester 4), long-term credits earned (Semester 7), on-time graduation (Semester 7), and graduation within nine semesters. Ability groups are defined by terciles of high school GPA in the pre-treatment cohorts. Points represent estimated coefficients relative to the 2015 cohort (t = 0), with vertical bars indicating 95 percent confidence intervals based on wild cluster bootstrap standard errors clustered at the program level. Negative values indicate outcomes below the reference cohort mean.

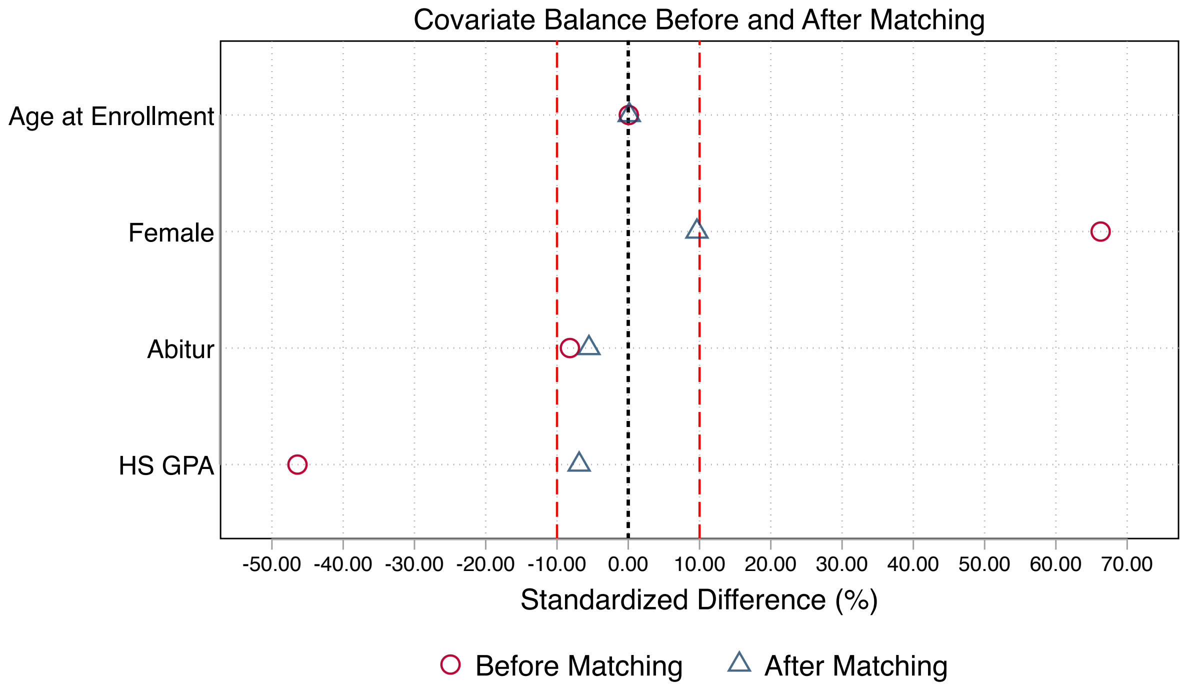

Figure A3. Standardized Covariate Differences Before and After Propensity Score Matching: Business Administration

Note: This figure displays standardized mean differences for selected covariates before and after propensity score matching. Circles represent unmatched (before matching) differences, and triangles represent matched (after matching) differences. Dashed vertical lines indicate conventional balance thresholds (±10%). Improved balance after matching is indicated by estimates closer to zero.

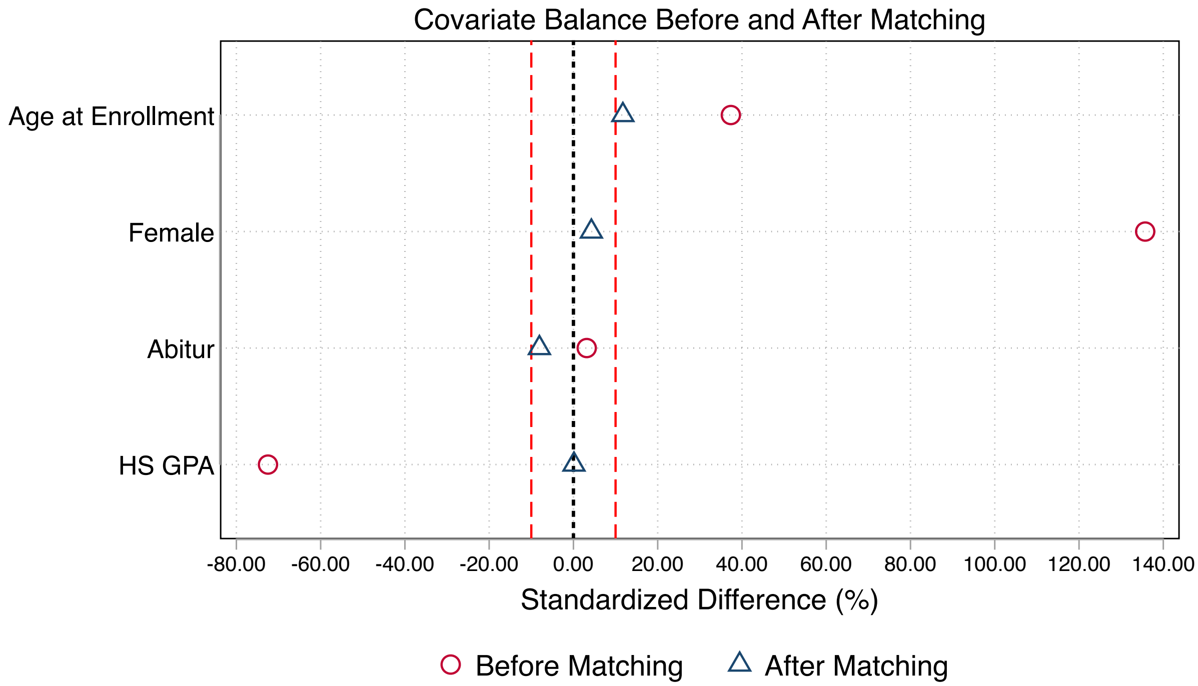

Figure A4. Standardized Covariate Differences Before and After Propensity Score Matching: SW

Note: This figure displays standardized mean differences for selected covariates before and after propensity score matching. Circles represent unmatched (before matching) differences, and triangles represent matched (after matching) differences. Dashed vertical lines indicate conventional balance thresholds (±10%). Improved balance after matching is indicated by estimates closer to zero.

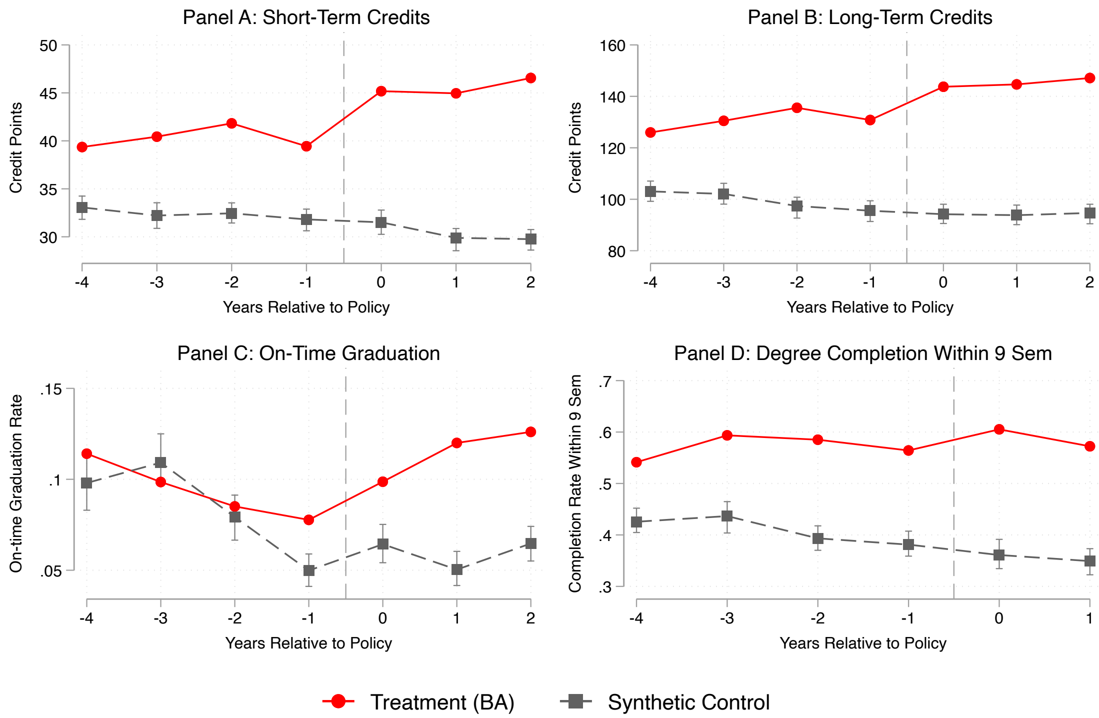

Figure A5: Synthetic Difference-in-Differences Trends in Business Administration. *Notes*: This figure plots the observed trajectories for the Business Administration program (solid red) and the synthetic counterfactual (dashed grey) using the Arkhangelsky et al. (2021) estimator with optimized unit and time weights. Panels A and B report cumulative credit accumulation at semesters 2 and 7, while Panels C and D display on-time graduation and degree completion within nine semesters.

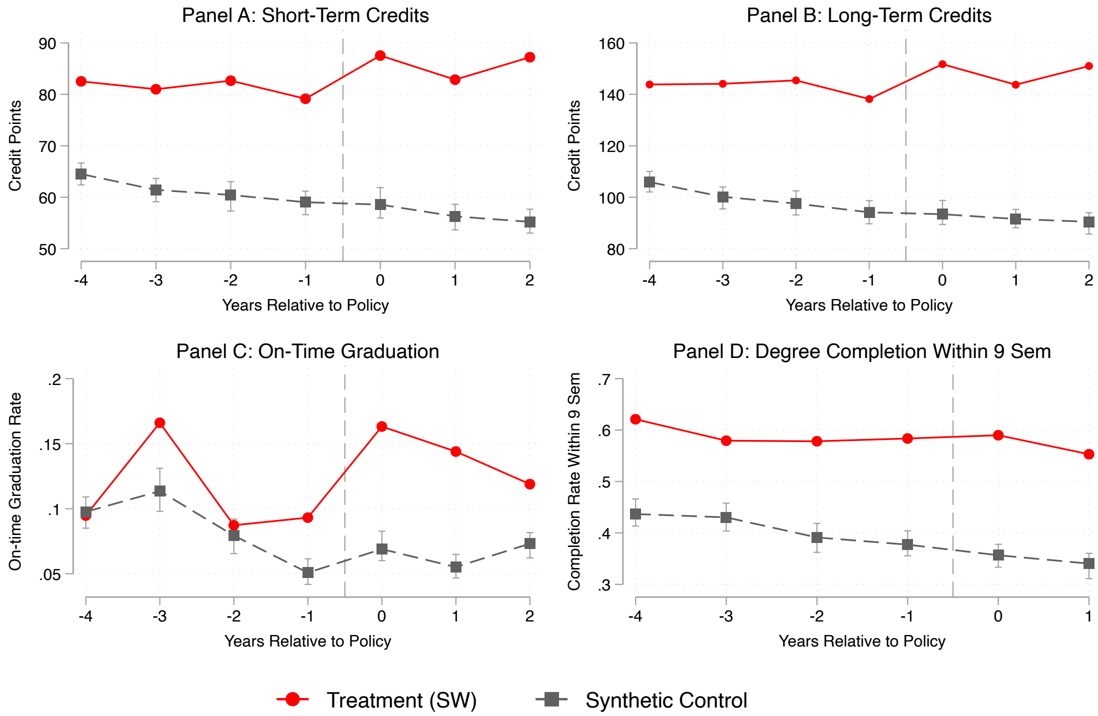

Figure A6: Synthetic Difference-in-Differences Trends in Social Work. *Notes*: This figure plots the observed trajectories for the Social Work program (solid red) and the synthetic counterfactual (dashed grey) using the Arkhangelsky et al. (2021) estimator with optimized unit and time weights. Panels A and B report cumulative credit accumulation at semesters 2 and 7, while Panels C and D display on-time graduation and degree completion within nine semesters.

Table A6. Summary of Regulatory Stability and Non-Deadline Changes

Category	Description of Change	Nature and Relevance
Panel A. Business Administration: Regulatory Adjustments		
Category	Description of Change	Nature and Relevance
Deadline Policy	Introduction of binding deadlines: Modules G1 (“General BWL”) and G2 (“Math”) must be passed by the end of Semester 1.	Primary treatment. The only binding progression constraint introduced.
Module Sizing	Variable module sizes (2–8 ECTS) standardized to a uniform 5-ECTS grid.	Administrative. Facilitates mobility and scheduling; total content volume unchanged.
Renaming	e.g., Einführung in die BWL → Allgemeine BWL; Materialwirtschaft → Logistik.	Cosmetic. Modernized titles; content unchanged.
Sequencing	Two modules (Macroeconomics, Business English) moved from Year 2 to Year 1.	Neutral. Balances workload; curriculum unchanged.
Specializations	Specialization requirement adjusted from 3×14 ECTS to 3×12 ECTS.	Technical. Credits redistributed; total degree remains 210 ECTS.
Panel B. Social Work: Regulatory Adjustments		
Deadline Policy	Introduction of binding deadline: All 60 first-year credits must be passed by the end of Semester 4.	Primary treatment. The only binding progression constraint introduced.
Renaming	e.g., Theorie/Geschichte → Einführung in die Wissenschaft; Theorien der SA → Wissenschaft der SA.	Cosmetic. Updated titles; content unchanged.
Thesis Registration	Registration deadline moved to “one month after start of 9th semester.”	Administrative. Aligns thesis timing with the new rules.
Assessment Forms	Removal of Referat (presentation) as mandatory partial exam in Modules 1.5 and 1.13.	Minor simplification. Reduced assessment complexity; workload unchanged.
Prerequisites	Removal of specific ECTS prerequisites for intermediate modules (e.g., Module 2.5).	Flexibility. Eliminates minor bottlenecks; smoother sequencing.

Notes: Based on the official Study and Examination Regulations (SPO) for Business Administration and Social Work. The 2016 reforms introduced binding course-completion deadlines as the only substantive change affecting student progression incentives. All other adjustments were administrative, cosmetic, or minor technical revisions that did not alter course content, credit requirements, or curricular structure.

Appendix B: Sensitivity to Parallel Trends Violations

A central concern in difference-in-differences estimation is the validity of the parallel trends assumption. While examining pre-treatment trends is standard practice, non-significant pre-treatment coefficients do not guarantee that parallel trends would have held in the post-treatment period. To rigorously assess the internal validity of the estimates, I implement the sensitivity analysis framework proposed by Rambachan and Roth (2023).

This method constructs "robust" confidence intervals by relaxing the strict parallel trends assumption. Instead of assuming that the difference in trends between treated and control groups is exactly zero in the post-treatment period, I allow for potential deviations from parallel trends. These deviations are bounded using the "Relative Magnitudes" restriction, which links future trend violations to observed pre-treatment dynamics.

Specifically, the sensitivity parameter $\bar{M}$ bounds the magnitude of post-treatment violations of parallel trends relative to the maximum deviation observed in the pre-treatment period. When $\bar{M} = 0$, the analysis imposes exact parallel trends as in standard OLS DiD; $\bar{M} = 1$allows post-treatment deviations to be as large as the largest pre-treatment violation; and values $\bar{M} > 1$ permit even larger deviations, thereby assessing robustness to substantial trend volatility. A higher Breakdown $\bar{M}$ implies that the estimated treatment effect is robust to larger violations of the identification assumption.

The Role of Alternative Estimators—The sensitivity analysis highlights that, while the reform's mechanism (increased credit accumulation) is robust across specifications, the downstream impact on graduation rates in the standard two-way fixed-effects framework is sensitive to the linearity of the trend assumptions. This finding underscores the importance of the alternative estimation strategies employed in the main text: Propensity Score Matching (PSM-DiD) and Synthetic Difference-in-Differences (SDID).

The sensitivity detected in the OLS estimates stems largely from the "equal weighting" of control units, which can allow units with poor pre-trend fits to influence the counterfactual. Both PSM-DiD and SDID address this directly by reweighting the control group to minimize pre-treatment trend differentials.

1. Synthetic DiD: By explicitly minimizing the pre-treatment prediction error, SDID constructs a counterfactual that adheres more strictly to the parallel trends assumption than the unweighted OLS specification. The fact that the graduation results remain positive and significant under SDID (as shown in Section 9) suggests that the "sensitivity" found in the OLS analysis is likely driven by control group heterogeneity rather than a fundamental absence of effect.

2. Matching: Similarly, PSM ensures that comparisons are made only between students with observable similarities, reducing the potential for time-varying confounders correlated with student characteristics.

The triangulation of evidence across standard OLS, Rambachan & Roth (2023) sensitivity analysis, and reweighting estimators (SDID/PSM) supports a causal interpretation of the findings. The sensitivity analysis confirms that the reform drove a robust, undeniable increase in study intensity (credits). While the standard OLS estimates for graduation rates are sensitive to trend assumptions, their persistence across reweighted estimators (which explicitly correct for trend imbalances) provides confidence that the extensive margin effects are not artifacts of selection or spurious trends. Thus, I interpret the sensitivity analysis not as evidence against the graduation effects, but as confirmation that proper control group construction is better achieved via the preferred SDID and matching specifications.

Table B1: Sensitivity Analysis (Business Administration)

Outcome	Time Period	Original Estimate [95% CI]	Breakdown $\bar{M}$	Robustness
Short-Term Credits	t=0	5.48*** [2.90, 8.06]	1.25	Robust
(Semester 2)	t=1	7.40*** [5.34, 9.46]	1.75	Highly Robust
	t=2	9.99*** [7.89, 12.09]	> 2.50	Highly Robust
Long-Term Credits	t=0	12.77*** [5.23, 20.30]	0.75	Moderate
(Semester 7)	t=1	16.14*** [8.62, 23.66]	0.75	Moderate
	t=2	22.48*** [14.10, 30.85]	1.25	Robust
On-Time Graduation	t=0	-0.000 [-0.04, 0.03]	N/A	N/A
	t=1	0.047** [0.007, 0.087]	0.25	Sensitive
	t=2	0.030 [-0.03, 0.09]	N/A	N/A
Graduation ≤ 9 Sem	t=0	0.042*** [0.014, 0.070]	0.25	Sensitive
	t=1	0.035** [0.004, 0.067]	0.25	Sensitive

Notes: The table reports 95% robust confidence intervals constructed using the method of Rambachan and Roth (2023). "Breakdown $\bar{M}$ " indicates the maximum value of $\bar{M}$ (relative to the maximal pre-treatment trend violation) for which the treatment effect remains statistically significant at the 5% level. N/A indicates the baseline estimate was not statistically significant; thus, no sensitivity analysis is available. Graduation ≤9 Semesters excludes the 2018 cohort. *** p<0.01, ** p<0.05, * p<0.1.* based on study program clustered standard errors.

Table B2: Sensitivity Analysis (Social Work)

Outcome	Time Period	Original Estimate [95% CI]	Breakdown $\bar{M}$	Robustness
Short-Term Credits	t=0	5.87** [1.26, 10.49]	0.25	Sensitive
(Semester 4)	t=1	5.81** [1.52, 10.10]	0.25	Sensitive
	t=2	12.56*** [7.54, 17.59]	0.75	Moderate
Long-Term Credits	t=0	10.54** [2.90, 18.19]	0.25	Sensitive
(Semester 7)	t=1	7.06 [-0.30, 14.43]	N/A	N/A
	t=2	19.01*** [10.62, 27.41]	1.25	Robust
On-Time Graduation	t=0	0.038* [0.003, 0.074]	0.25	Sensitive
	t=1	0.050** [0.011, 0.090]	0.25	Sensitive
	t=2	0.005 [-0.056, 0.065]	N/A	N/A
Graduation ≤ 9 Sem	t=0	-0.013 [-0.041, 0.016]	N/A	N/A
	t=1	-0.011 [-0.040, 0.019]	N/A	N/A

Notes: The table reports 95% robust confidence intervals constructed using the method of Rambachan and Roth (2023). "Breakdown $\bar{M}$ " indicates the maximum value of $\bar{M}$ (relative to the maximal pre-treatment trend violation) for which the treatment effect remains statistically significant at the 5% level. N/A indicates the baseline estimate was not statistically significant; thus, no sensitivity analysis is available. Graduation ≤9 Semesters excludes the 2018 cohort. *** p<0.01, ** p<0.05, * p<0.1.* based on study program clustered standard errors.